



Contents lists available at ScienceDirect

Journal of Empirical Finance

journal homepage: www.elsevier.com/locate/jempfinThe decay of cay ☆Moritz Dauber^{ID}*, Jochen Lawrenz^{ID}

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ARTICLE INFO

Keywords:

Consumption–wealth ratio
Intertemporal budget constraint
Cointegration
Predictive regression
Inequality

ABSTRACT

We revisit the ability of the consumption–wealth ratio (cay) to forecast stock market returns and document a substantial decline in predictability over the last two decades. This decay of cay goes along with a structural shift in the underlying cointegration relationship, which can be attributed to the fact that asset wealth evolves increasingly detached from aggregate consumption and labor income. We propose a new version of cay derived only from the top 10% richest households and show that among various other proposed improvements of cay , this appears as the most promising empirical proxy for the still appealing theory.

1. Introduction

A basic tenet of neoclassical financial economics is that the representative agent maximizes utility over consumption under some budget constraint. Following this logic and building on ideas in Campbell and Mankiw (1989), Lettau and Ludvigson (2001) exploit the economic relations of the intertemporal budget constraint between wealth, consumption and expected returns to develop the consumption–wealth ratio (cay) as a predictor for stock market excess returns in the time-series. However, although being theoretically appealing, it has received ambiguous empirical support.

In this paper, we revisit the forecasting ability of cay and offer three contributions: *First*, we conduct comprehensive in-sample, out-of-sample, as well as economic significance tests on U.S. post-war data and document a remarkable and sustained decline of the predictive power of cay during (at least) the last decade. While prior literature has already documented poor out-of-sample performance, our results show that from the perspective of most recent data, cay appears to have lost even its in-sample ability to predict stock market returns. We conclude that, independent of how to assess predictability, the most recent evidence suggests that cay does not reliably predict stock market returns, and therefore lends strong support to earlier contributions, such as, e.g., Avramov (2002), Welch and Goyal (2008), and more recently (Goyal et al., 2024), who all voice concern about the predictive performance of cay . Our results are most salient for the original construction of cay from NDS (i.e., expenditures on nondurables and services) consumption data. Constructing cay from PCE (i.e., total personal consumption expenditures) data instead, as recently suggested in Lettau and Ludvigson (2019),¹ shows better performance but features the same decay up to the point of being currently only marginally significant at best.

Having documented the substantial decay of cay , the focus of our contribution is on investigating econometric as well as economic reasons for the puzzling discrepancy between the poor empirical support and the a priori compelling theoretical justification. Thus,

☆ We are grateful to Otto Randl (discussant), Konark Saxena (discussant), Dennis Umlandt, and all participants at the 2023 Forecasting Financial Markets Conference, 2023 AWG Workshop, 2023 IDS Workshop, 2023 Computational and Financial Econometrics Conference, 2023 Doctoral Seminar Obergurgl, 2024 Doctoral Seminar Liechtenstein, 2024 Annual Meeting of the International Association for Applied Econometrics and the 2024 Annual Meeting of the European Financial Management Association.

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¹ Lettau and Ludvigson (2019) recognize that NDS data as a fraction of total consumption declined substantially over the past decades since it ignores the service flow from durables, which grew relatively more important.

<https://doi.org/10.1016/j.jempfin.2025.101668>

Received 9 December 2024; Received in revised form 21 August 2025; Accepted 19 November 2025

Available online 28 November 2025

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our *second* contribution is to uncover that the declining predictive power of *cay* goes along with a structural shift in its presumed cointegration relationship. We find that the coefficient on wealth in the cointegrating vector shows a steady decline over at least the last two decades, both in terms of value and significance up to the point that it ends up insignificant from the perspective of most recent data. Furthermore, the econometric evidence to reject the null of no cointegration has weakened substantially, so that there is currently no reason to believe that a stable cointegration relation exists in the first place. From a purely econometric point of view, we argue that these structural shifts are largely responsible for the *decay* in the forecasting ability of *cay* and can also explain the strong discrepancy between the in-sample (IS) and (strict) out-of-sample (OOS) performance of *cay*.

Our *third* contribution is to investigate different (economic) avenues how to reconcile the (rather disappointing) empirical evidence with the (still appealing) theory. Recall that *cay* is supposed to be predictive due to the optimizing behavior of the representative agent who adjusts consumption to expected wealth changes. Indeed, [Christelis et al. \(2015\)](#) do find evidence from detailed survey data, that U.S. households adjust their expenditures in response to wealth shocks, in particular, when they perceive them to be permanent.

However, against the background of significant and further rising wealth and income inequalities, it is questionable if market aggregates are able to proxy for the behavior of the representative investor. Thus, inspired by recent work of [Lettau et al. \(2019\)](#), who propose capital share risk as a surprisingly successful proxy for the pricing kernel,² we construct a novel measure of *cay* from the data of the 10% richest U.S. households (labeled as $cay^{NDS\ 10}$ and $cay^{PCE\ 10}$). Given that rich households tend to hold a disproportionately large fraction of stock market wealth relative to their consumption, we hypothesize that $cay^{PCE\ 10}$ should be a stronger predictor if the theory is appropriate.³

We contrast our novel $cay^{PCE\ 10}$ measure with the original *cay* of [Lettau and Ludvigson \(2001\)](#) as well as different alternatives to *cay* that have been put forward in the recent literature to construct better proxies for the underlying theory, such as disentangling financial wealth as in [Sousa \(2010\)](#), accounting for omitted variables as in [Guo \(2006\)](#), or unfiltering consumption data in the spirit of [Kroencke \(2017\)](#). Although we cannot entirely mitigate the decay in predictive ability, it appears to be way less pronounced. In fact, we find that $cay^{PCE\ 10}$ turns out as the dominant predictor in-sample as well as in terms of economic performance. We interpret this finding as support for the more general conclusion that the avenue for reconciling the (rather disappointing) empirical evidence with the (still appealing) theory is to identify more properly the consumption and wealth data of the representative investor by taking into account the significant wealth and income inequalities.

In our analysis, we conduct in-sample, out-of-sample and economic performance tests with U.S. post-war data from 1952:Q1 up to 2019:Q4. We deliberately end our sample for baseline results before the Covid-19 episode, since pandemic-induced shutdowns are hard to reconcile with the underlying theory of *cay*. An analysis that includes the Covid-19 episode (i.e., including data up to 2022:Q4) is provided separately in an [Appendix A](#). We show that, independent of the way how to assess *cay*, it shows a remarkable decline over at least the last decade to end up insignificant from the perspective of IS, OOS and economic performance. [Fig. 1](#) illustrates a preview on one of our main points. In panel B, we show (in-sample) predictive regression coefficients for the original *cay*-measure (in black) and its top-10% PCE-counterpart $cay^{PCE\ 10}$ (in blue). Beginning before the global financial crisis, the standard measure starts declining to end up insignificant from the perspective of 2019:Q4. To the contrary, the forecasting ability of the consumption-wealth ratio for the richest 10% of households remains remarkably stable and significant. The reason for this substantial decay in the original measure can be seen in panel A of [Fig. 1](#). It displays the input data to the construction of *cay* which is the log of consumption (NDS as well as PCE), asset wealth and labor income after removing a deterministic time trend. We do observe a strong comovement between consumption and income over the entire sample. In contrast, asset wealth displays a strongly detached development from consumption and income, which is particularly pronounced since around 2000, and manifests itself as a structural shift in the cointegration relationship where the coefficient on asset wealth declines steadily. Taking the claim of [Lettau and Ludvigson \(2005b\)](#) to use the most comprehensive (i.e., most recent) data set for the estimation of the equilibrium at face value, leaves us with a time series for *cay* that would have never displayed forecasting power neither in- nor out-of-sample.

Our assessment of the predictive power of *cay* is clearly in contrast to the original work of [Lettau and Ludvigson \(2001\)](#) as well as their update in [Lettau and Ludvigson \(2019\)](#). It is also in contrast to the earlier works of [Hahn and Lee \(2006\)](#), [Guo \(2006\)](#), [Kalotay et al. \(2007\)](#), [Della Corte et al. \(2010\)](#), [Guo et al. \(2013\)](#), and [Ren et al. \(2014\)](#) who concede *cay* as having predictive power. Our results rather lend support to [Avramov \(2002\)](#), [Welch and Goyal \(2008\)](#), and the most recent assessment of [Goyal et al. \(2024\)](#) which are skeptical or outright negative about the forecasting ability of *cay*.⁴ The conflicting results in this prior literature are largely due to diverging opinions with respect to underlying assumptions of the cointegration relationship and how to conduct OOS tests.⁵ Therefore, it is instructive to analyze in more detail the origin of the discrepancy of earlier results to our current assessment.

² Note also the recent discussion and critique in [Gospodinov and Robotti \(2021\)](#).

³ [Lustig et al. \(2013\)](#), e.g., estimate that in the market aggregate, stock market wealth represents only 1% of total wealth. It is reasonable to expect that among rich households, this fraction will be substantially larger.

⁴ [Goyal et al. \(2024\)](#) assess a large number of predictors and do not analyze or discuss their negative evidence for *cay* in any detail.

⁵ [Avramov \(2002\)](#) and later ([Welch and Goyal, 2008](#)) argue that the construction of *cay* suffers from a 'look-ahead bias' when the cointegrating vector is estimated using the full sample and point towards the weak out-of-sample performance when the test is carried out under the assumption that only available information is used for prediction. [Brennan and Xia \(2005\)](#) address the same 'look-ahead bias' and argue that a cointegration relationship between wealth and a simple time trend would generate similar (or even better) forecasting results. The conflicting results are due to different conceptions of how to conduct OOS tests for the residual of a cointegration relationship, and more generally due to the substantial model uncertainty for cointegration test as pointed out by, e.g., [Koop et al. \(2008\)](#). While ([Avramov, 2002](#)), [Welch and Goyal \(2008\)](#), and [Brennan and Xia \(2005\)](#) favor the perspective of a real-time decision-maker and only allow for contemporaneous available information, [Lettau and Ludvigson \(2001\)](#) and more explicitly ([Lettau and Ludvigson, 2005b](#)) do reply to this criticism by making the strong case, that using only partial information to estimate the equilibrium cointegrating vector is inappropriate because it neglects information that would have been available to the decision-maker if the underlying theory is correct.

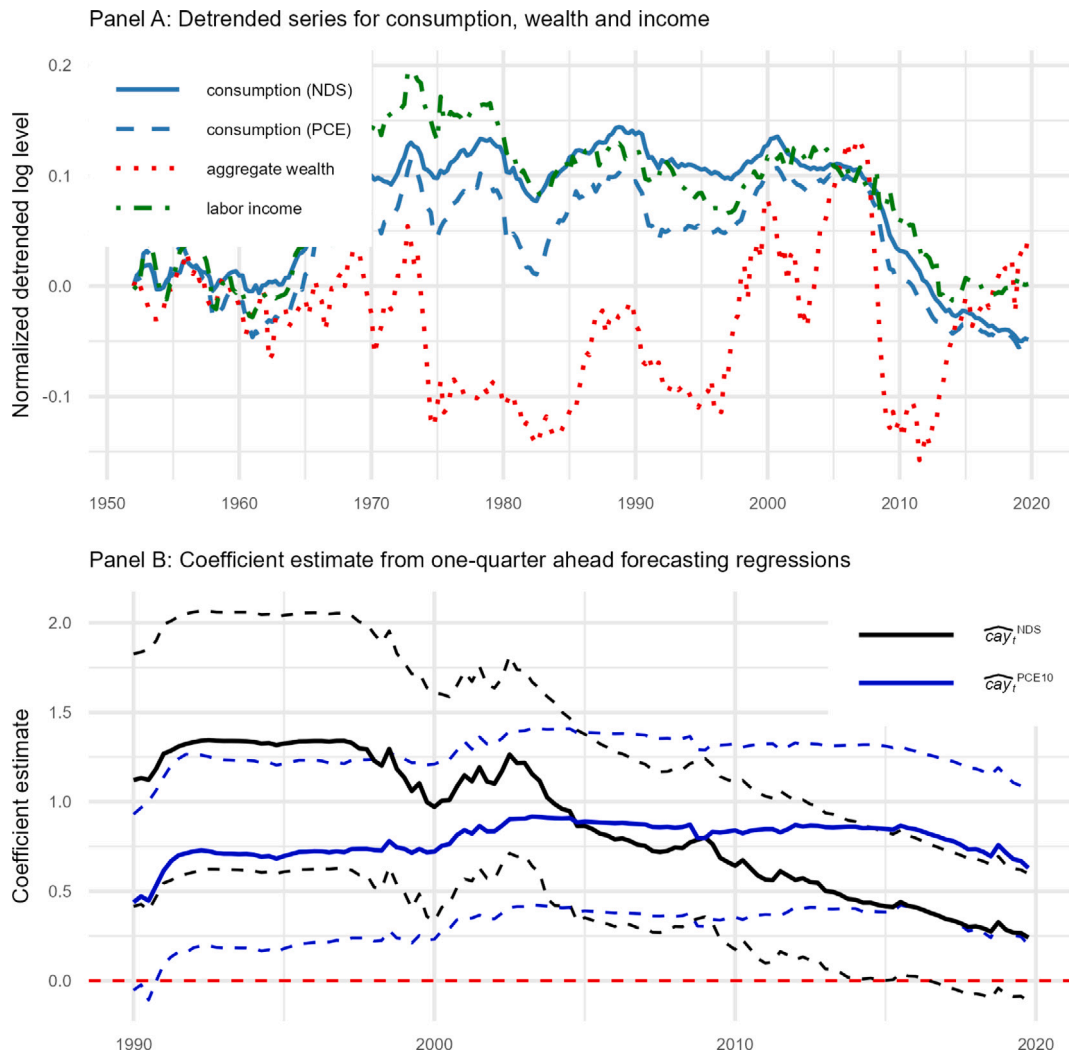


Fig. 1. Panel A shows NDS consumption (solid blue), PCE consumption (dashed blue), aggregate wealth (red) and labor income (green) in normalized detrended log levels. This means that the log series are detrended assuming a simple linear time trend and afterwards are normalized to start at zero.

The lower panel shows the coefficient estimates of $\widehat{cay}_t^{NDS}$ (solid black) and $\widehat{cay}_t^{PCE10}$ (solid blue) from quarterly forecasts of excess returns on the CRSP NYSE/NYSE MKT/NASDAQ/Arca Value-Weighted Market Index, accompanied by their 90%-confidence bands as the respective dashed lines. For more details see Section 3. The sample period is 1952:1–2019:4.

This analysis focuses on the stability of the cointegration relationship and shows that – as mentioned above – sustained shifts have taken place, in particular with respect to the coefficients of the cointegrating vector. A related aspect is the role of a deterministic time trend in the cointegration equation. Brennan and Xia (2005) and later (Hahn and Lee, 2006) point towards the potentially critical role of an omitted time trend. In particular, Hahn and Lee (2006) show that the performance of cay is biased upwards when the cointegrating relationship is restricted to have no time trend. Although substantially smaller, their results still find forecasting power for cay when estimated with a time trend. Our findings are even stronger, in that cay is actually uninformative over the entire sample when estimated with a time trend. However, we point out that from the perspective of most recent data, the inclusion of a time trend is econometrically no longer justified and rather confirms the original assumption of Lettau and Ludvigson (2001). In this sense, we document some irony of history. Up until around 2010, the original cay measure displayed stable forecasting power, but was subject to the potential criticism that it earned its predictability due to the omitted (but statistically significant) time trend. Since 2010, the data does support the assumption of omitting a time trend, but cay no longer displays forecasting power.

A further implication of the sustained shift in the cointegrating vector is the fact that cay tends to be always biased downwards in the most recent observations, which follows from the fact that the fixed coefficient estimates are estimated over the full sample and will therefore overweight asset wealth in the latest periods. This finding is able to explain the strong discrepancy between the in-sample and out-of-sample performance results of cay .

We complement our main analysis with comprehensive robustness checks. Among others, we test various cointegration regression approaches. We also pick up the argument by Lettau and Ludvigson (2005a,b) to avoid estimating the cointegrating relationship and rather conduct a multivariate regression. None of these tests change our main conclusions that the last decades have seen a sustained decay of *cay* and that the way forward to reconcile empirics with the theory is to account for increasing wealth inequalities.

The remainder of the article is structured as follows: Section 2 recaps the economic theory behind the consumption–wealth ratio and the estimation of *cay*. It also develops the version of *cay* for the top 10% richest households and discusses the choice of the employed consumption series. Section 3 documents the decay of *cay* from an in-sample, out-of-sample and economic perspective. In Section 4, we address the structural shift in the cointegration relationship, including the role of a deterministic time trend. Section 5 considers various other attempts to enhance the predictability of *cay*. Section 6 discusses a battery of robustness checks before Section 7 concludes.

2. Methodology

2.1. The consumption–wealth ratio

Starting out with an agent's intertemporal budget constraint

$$W_{t+1} = (1 + R_{w,t+1}) (W_t - C_t), \quad (1)$$

where W_t denotes aggregate wealth, C_t denotes aggregate consumption and $R_{w,t+1}$ is the return on aggregate wealth between periods t and $t + 1$, Campbell and Mankiw (1989) derive the log consumption–wealth ratio as

$$c_t - w_t \approx E_t \sum_{i=1}^{\infty} \rho^i (r_{w,t+i} - \Delta c_{t+i}). \quad (2)$$

This means, the consumption–wealth ratio is a function of future expected returns and future expected consumption growth and must thus forecast either one of the two, or both. This crucial linkage allows us to infer expectations on future returns from the consumption–wealth ratio using observable consumption data.

The consumption–wealth ratio, however, is not directly observable since we are not able to observe one of the components of total wealth, namely human capital, H_t . Lettau and Ludvigson (2001) resolve this issue by using log labor income, y_t , as a proxy for the nonstationary component of log human capital, h_t , by approximating

$$w_t \approx \alpha a_t + (1 - \alpha) h_t \quad (3)$$

where a_t denotes log asset wealth and α represents the average share of asset holdings in total wealth, i.e., A/W . They then replace h_t with $\kappa + y_t + z_t$ where κ denotes an unimportant constant and z_t is a mean zero stationary random variable. Together with the approximation

$$r_{w,t} \approx \alpha r_{a,t} + (1 - \alpha) r_{h,t} \quad (4)$$

for log returns by Campbell (1996), this transforms (2) into

$$c_t - \alpha a_t - (1 - \alpha) y_t = E_t \sum_{i=1}^{\infty} \rho^i ([\alpha r_{a,t+i} + (1 - \alpha) r_{h,t+i}] - \Delta c_{t+i}) + (1 - \alpha) z_t. \quad (5)$$

Given that all terms on the right-hand side are assumed to be stationary, the left-hand side must also be stationary, implying that c_t , a_t and y_t must be cointegrated. Hence, the left-hand side represents the deviation in the common trend between consumption, aggregate wealth, and labor income, which, according to Lettau and Ludvigson (2001), defines *cay* as

$$cay_t = c_t - \alpha a_t - (1 - \alpha) y_t. \quad (6)$$

From (5), we get that as long as expected future returns on human capital, $r_{h,t+i}$, and consumption growth, Δc_{t+i} , are not too variable, or at least highly correlated with expected returns on assets, $r_{a,t+i}$, the consumption–wealth ratio proxied for by *cay*, should predict market expectations on future asset returns.

2.2. The consumption–wealth ratio of the rich

The basic rationale behind *cay* is derived from the representative agent's intertemporal budget constraint, cf. (1). Based on expectations of future returns, the agent smooths her contemporaneous consumption accordingly. In the standard empirical implementation, *cay* is estimated from aggregate consumption flow and aggregate wealth. While in theory this aggregation seems most natural, in practice, the distribution of (asset) wealth is strongly unequal. Lettau et al. (2019) report that the top 10% of households along the wealth distribution own more than 70% of stock wealth among stockowners, and even around 85% among all households. This is partly due to limited capital market participation, e.g., Lettau et al. (2019) also report that while the raw stock market participation rate has increased from 31.7% in 1989 to a more or less constant level of around 50% since the end of the 1990s, the wealth-weighted participation rate remains at far lower levels of only 20.2% in 2013. These findings seem to not

have changed substantially until the end of our sample in 2019 as recently pointed out in a Wall Street Journal article.⁶ Changes thereafter, evolving from the pandemic and zero-commission trading, therefore do not impact our findings. We can thus conclude that a substantial part of total consumption is contributed from households that do not own any sizable financial wealth and consequently their intertemporal budget constraint should not materially be affected by expectations over future capital market returns.⁷

This empirical fact leads us to consider a refined version of the consumption–wealth ratio that attempts to only account for the consumption and wealth of the top 10% richest households in order to (re-)strengthen the theoretical rationale behind *cay*. Guided by the recent work of Lettau et al. (2019), who propose short- and medium-frequency fluctuations in the capital share as a surprisingly successful proxy for the pricing kernel, we construct adjusted series of consumption, asset wealth and labor income for the 10% richest households. We do so by regressing the income and wealth share of the top 10% richest households onto the capital share and in a second step multiply the fitted values for the income share with the original aggregate consumption and labor income series and the fitted values for the wealth share with the original asset wealth series. Naturally, this is done in levels before converting back to log series.

The two regressions of the income and wealth share on the capital share are described by

$$AS_t^{\text{top10}} = \alpha_A + \beta_A KS_t + \varepsilon_t, \quad (7)$$

$$YS_t^{\text{top10}} = \alpha_Y + \beta_Y KS_t + \varepsilon_t, \quad (8)$$

where AS_t^{top10} and YS_t^{top10} denote the wealth and income share of the top 10% richest households, respectively, and KS_t is the capital share, which is defined as $1 - LS_t$ where LS_t is the labor share. The fitted values $\widehat{AS}_t^{\text{top10}}$ and $\widehat{YS}_t^{\text{top10}}$ are multiplied by the aggregate income, wealth and consumption time series to obtain: $C_t^{\text{top10}} = C_t \cdot \widehat{YS}_t^{\text{top10}}$, $Y_t^{\text{top10}} = Y_t \cdot \widehat{YS}_t^{\text{top10}}$, and $A_t^{\text{top10}} = A_t \cdot \widehat{AS}_t^{\text{top10}}$. After transforming back to logs, our top-10 *cay* version is obtained as

$$cay_t^{\text{top10}} = c_t^{\text{top10}} - \alpha^{\text{top10}} a_t^{\text{top10}} - (1 - \alpha^{\text{top10}}) y_t^{\text{top10}} \quad (9)$$

with α^{top10} being the average share of asset wealth in total wealth among the top-10% richest households.

We will refer to this version of *cay* as the *top-10 specification*, in contrast to the *aggregate specification* in (6) which uses the aggregate input series. As source for the inequality measures, we resort to the work of Saez and Zucman (2016). Although their original dataset ends in 2014, Gabriel Zucman himself notes on his website, that the series are continued within the World Inequality Database (WID), which we use to get AS_t^{top10} and YS_t^{top10} . The WID data is widely used but also not uncontested (see, e.g., the discussion between (Saez and Zucman, 2022) and Smith et al. (2023)). A voluminous strand in the economics literature discusses theoretical and empirical issues in the measurement of inequality. Alternative datasets are, e.g., available from (Kopczuk and Saez, 2004), Kuhn et al. (2020), or (Smith et al., 2023) and put forward substantially different estimates. Gomez (2025) concludes that there is tremendous uncertainty in measuring the evolution of wealth shares. We do not intend to take any stance in this discussion, so in an attempt to construct a reasonable proxy of the wealth and income share, we resort to the methodology put forward in Lettau et al. (2019) to construct them from fitted values in the above regression on the capital share, KS_t . We report estimation results from different data sources in Panel A of Table A.1 of Appendix A.⁸ The same Appendix A also contains further additional information on the implications of using fitted values. Among others, we show that the projection on KS_t does not mechanically increase the stability of the cointegration relation in eq. (9).

Note that the theoretical basis of *cay* (as outlined in the previous section) will still be valid for cay_t^{top10} as long as the return on the aggregate wealth portfolio (R_w) is sufficiently strongly correlated with the wealth of the top-10% richest households. Or in other words, if the richest households also tend to be the marginal investors. There is indeed evidence supporting this assumption by Lettau et al. (2019), who conjecture that the stochastic discount factor is driven solely by marginal utility of consumption by rich households. Within Section 3. (Tab.VIII), they present empirical evidence for this claim and conclude that “*rich stockowners’ consumption growth strongly explain(s) return premia on all equity portfolios*” (Lettau et al., 2019, p. 1789).

We take this evidence together with the above mentioned descriptive statistics of stock wealth distribution, as justification for our hypothesis that the top-10 specification of the consumption–wealth ratio should capture the underlying theory more accurately and in consequence produce stronger forecasting results. We show that this is indeed the case in Section 3.

To be clear, with our top-10 specification, we put forward an improvement in the construction of an empirical proxy for *cay*. Developing a full-fledged theoretical derivation of the top-10 *cay* as a predictor is beyond the scope of the current paper.

⁶ The article can be found under <https://www.wsj.com/finance/stocks/stocks-americans-own-most-ever-9f6fd963>.

⁷ By using consumer expenditure survey data, Smoluk and Bennett (2008) find that consumption adjusts to anticipated housing wealth only among richer households and not for the lowest income quintile.

⁸ Note, that the original series are on an annual basis. Since the capital share series is quarterly we take yearly averages for the capital share in order to run regressions (7) and (8). To obtain the fitted values for the wealth and income share, we revert to the original quarterly capital share series together with the estimated coefficients from (7) and (8).

2.3. Estimation of the cointegration relation

Under the assumption of the existence of a (single) cointegration relationship between consumption, aggregate wealth and labor income, the goal is to obtain an estimate for α from eq. (6). We will address the actual validity of this fundamental assumption in all detail in Section 4. For now, we simply take it as given.

We follow Lettau and Ludvigson (2001) in employing Stock and Watson's (1993) dynamic least squares (DLS) technique that adds leads and lags of the first differences in aggregate wealth and labor income to an otherwise normal ordinary least squares (OLS) regression of consumption on aggregate wealth and labor income. The number of leads and lags included in the regression does not make a difference for numbers between one and eight, which is why we do not depart from Lettau and Ludvigson (2001) in choosing a lead/lag length of eight to estimate

$$c_t = \alpha + \beta_a a_t + \beta_y y_t + \sum_{i=-8}^8 b_{a,i} \Delta a_{t-i} + \sum_{i=-8}^8 b_{y,i} \Delta y_{t-i} + \varepsilon_t \quad (10)$$

where Δ denotes the difference operator.

The main advantage of a DLS specification over a simple OLS specification is that it eliminates the effects of regressor endogeneity. Despite that, Lettau and Ludvigson (2001) point out that the estimates of β_a and β_y will be consistent, even though the error term ε_t will typically be correlated with a_t and y_t . This comes from the fact that OLS estimates of cointegrating parameters converge to the true parameters at a faster rate than in normal estimations (Stock, 1987).

With the estimated parameters $\hat{\beta}_a$ and $\hat{\beta}_y$ from (10), we then define

$$\widehat{cay}_t = c_t - \hat{\beta}_a a_t - \hat{\beta}_y y_t. \quad (11)$$

This estimated $\widehat{cay}$ can be thought of as the 'residual' from the cointegration regression and therefore represents an estimate for the trend deviation in the long-run relationship between consumption, aggregate wealth and labor income, i.e., a proxy for the consumption-wealth ratio.

2.4. Measure of consumption

A crucial part in estimating $\widehat{cay}$ as in (11) is obviously the choice of which consumption measure to employ. We follow two different approaches and use this subsection to outline the reasoning behind this decision.

Our first measure of consumption is the typically employed nondurables and services consumption excluding clothing and footwear (NDS) from the NIPA tables by the BEA. The idea behind using NDS instead of total personal consumption expenditures (PCE) is, that the former more accurately captures the flow of consumption, whereas expenditures on durable goods that are contained in PCE are rather to be seen as additions and replacements to existing stock. Lettau and Ludvigson (2001) assume that total consumption, i.e., PCE, is a constant multiple of NDS. It allows them to assume a constant scale factor between log total consumption and log NDS, leading to a valid cointegration relationship between NDS consumption, aggregate wealth and labor income.

In light of recent developments, we also pick up a change of methodology outlined in Lettau and Ludvigson (2019) that stems from the fact that the assumption of a constant scale factor between NDS and PCE seems no longer valid. In lack of any other suitable measure for the flow of consumption, Lettau and Ludvigson (2019) conclude that, under certain assumptions, PCE is the more appropriate consumption measure. Interestingly, Lettau and Ludvigson (2019) make another case for PCE instead of NDS by arguing that it better captures the consumption of particularly wealthy households as they consume disproportionately large amounts of durable goods. We address this issue more directly with our top-10 specification introduced above but switching from NDS to PCE we would expect said specification to perform even better. Again, we show that this is the case in Section 3.

In total, we therefore consider two baseline versions of $\widehat{cay}$, denoted as $\widehat{cay}_t^{\text{NDS}}$ and $\widehat{cay}_t^{\text{PCE}}$ and refer to them as the aggregate specifications. Their adjusted counterparts that capture only the consumption-wealth ratio of the top 10% richest households are denoted by $\widehat{cay}_t^{\text{NDS}10}$ and $\widehat{cay}_t^{\text{PCE}10}$.

2.5. Data

Before estimating $\widehat{cay}_t$, we briefly describe our data. A detailed description of how each series is derived can be found in Appendix B. Unless otherwise noted, we use quarterly samples spanning the period of 1952:1–2019:4.

All data on consumption and income are taken from the NIPA tables by the Bureau of Economic Analysis (BEA). The wealth series come from the Financial Accounts publication by the Board of Governors of the Federal Reserve System (FED).

The deflators used to transform each raw consumption series (e.g., PCE, nondurables or services) into real terms also come from the BEA. The asset wealth and income series are deflated with the PCE deflator.

The data on income and wealth inequality come from the World Inequality Database (WID) and are taken from the U.S. Distributional National Accounts.

Following Lettau et al. (2019), our measure for the labor share is the labor share of the nonfarm business sector by the Bureau of Labor Statistics (BLS).

Asset returns are calculated with the CRSP NYSE/NYSE MKT/NASDAQ/Arca Value-Weighted Market Index provided by the Center for Research in Security Prices (CRSP). We refer to this index as the market portfolio. Interest rates are collected from the Board of Governors of the Federal Reserve System's H.15 Selected Interest Rates publication. Our proxy for the risk-free rate is the 3-Month Treasury Bill Secondary Market Rate.

Table 1

Cointegrating parameter estimates for the baseline and top-10 specification for both consumption measures. Newey and West (1987) corrected t -statistics appear in parentheses. The sample period is 1952:1–2019:4.

	aggregate		top-10%	
	NDS	PCE	NDS	PCE
$\hat{\beta}_a$	0.029 (0.809)	0.235*** (8.148)	0.091** (2.081)	0.280*** (9.981)
$\hat{\beta}_y$	0.929*** (24.243)	0.786*** (25.072)	0.874*** (19.306)	0.766*** (27.810)

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 2

Summary statistics for $\widehat{cay}_t^{\text{NDS}}$, $\widehat{cay}_t^{\text{PCE}}$, $\widehat{cay}_t^{\text{NDS } 10}$, $\widehat{cay}_t^{\text{PCE } 10}$ and quarterly excess returns. The sample period is 1952:1–2019:4.

	$\widehat{cay}_t^{\text{NDS}}$	$\widehat{cay}_t^{\text{PCE}}$	$\widehat{cay}_t^{\text{NDS } 10}$	$\widehat{cay}_t^{\text{PCE } 10}$	$r_t - r_{f,t}$
Panel A: Univariate summary statistics					
Mean	0.487	−0.497	0.195	−0.948	0.015
Standard deviation	0.024	0.020	0.023	0.021	0.083
Autocorrelation	0.948	0.907	0.936	0.880	0.065
Panel B: Correlation matrix					
$\widehat{cay}_t^{\text{NDS}}$	1.000	0.837	0.963	0.651	0.053
$\widehat{cay}_t^{\text{PCE}}$		1.000	0.868	0.918	−0.035
$\widehat{cay}_t^{\text{NDS } 10}$			1.000	0.771	0.056
$\widehat{cay}_t^{\text{PCE } 10}$				1.000	0.001
$r_t - r_{f,t}$					1.000

3. Empirical analysis

This section documents first the results for the cointegration relation, followed by predictive regression estimates. We then show results for the in-sample, out-of-sample and economic performance tests. Throughout this section, we report results for the traditional cay measure (both from NDS and PCE consumption) and contrast it with our novel top-10 approach $\widehat{cay}_t^{\text{NDS } 10}$ and $\widehat{cay}_t^{\text{PCE } 10}$. A comparison with alternative cay -specifications is provided in Section 5.

3.1. Estimation of the cointegration relation

In a first step, we estimate the cointegration regression (10) for the two aggregate and the two top-10 specifications. The results are reported in Table 1. Next to using the DLS estimates throughout the paper, we also report the cointegrating parameters for other estimation methods, namely OLS, Park's (1992) canonical cointegration regression (CCR) and Phillips and Ouliaris' (1990) fully modified estimator (FME) in Appendix C.1.

While the coefficient estimate for labor income is highly significant in all four regressions, the coefficient estimate of asset wealth is not significant in the aggregate specification and only weakly significant in the top-10 specification in the case of NDS. While for PCE, the coefficient on asset wealth is highly significant for both specifications. We will later see in Section 4 that this is the most recent manifestation of an ongoing shift in the underlying cointegration relation that has been going on for the last two decades since the inception of cay in Lettau and Ludvigson (2001).

Our estimates for the cointegrating parameters from Table 1 now allow us to estimate cay according to (11). Summary statistics, including the log excess return of the market portfolio, $r_t - r_{f,t}$, where r_t denotes the log return of the market portfolio and $r_{f,t}$ denotes the log risk-free rate, can be found in Table 2.

Fig. 2 shows the aggregate cay using NDS and the top-10 cay using PCE in units of standard deviation, together with log excess returns and NBER recessions. In general, we see that in the past, negative developments in cay often foreshadowed negative excess returns, e.g., the oil crisis in the 1970s or the bear market after the burst of the dotcom bubble in the early 2000s. Other events, such as the 1987 stock market crash, but also the financial crisis of 2007/08, however, were not accompanied by a foregone decline in cay . More recently, cay has entered an ongoing negative development since the financial crisis, albeit mostly positive returns during the last decade. Notably, this finding holds independent of the specification. The negative development of cay during especially the last decade is in sharp contrast to the positive stock market development in that time span. This observation already serves as a preview into the declining forecasting ability of the consumption–wealth ratio, that we address in the following in more detail.

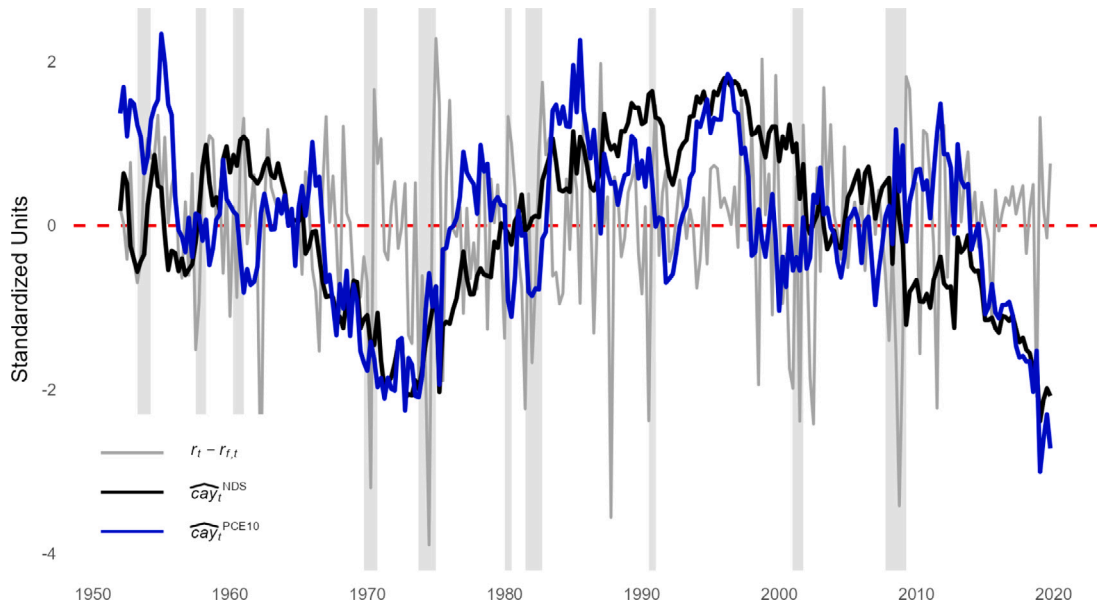


Fig. 2. The plot shows $\widehat{cay}_t^{\text{NDS}}$ as the black and $\widehat{cay}_t^{\text{PCE10}}$ as the blue line. The gray curve represents log excess returns, $r_t - r_{f,t}$. The gray shaded areas mark NBER recessions. The sample period is 1952:1–2019:4.

Table 3

The table reports forecasting regression estimates for $\widehat{cay}_t^{\text{NDS}}$, $\widehat{cay}_t^{\text{PCE}}$, $\widehat{cay}_t^{\text{NDS10}}$ and $\widehat{cay}_t^{\text{PCE10}}$. Forecasts span different horizons from one quarter up to five years. Newey and West (1987) corrected t -statistics appear in parentheses and adjusted R^2 statistics in square brackets. The sample period is 1952:1–2019:4.

	Dependent variable: $\tilde{r}_{t,H}$						
	$H = 1$	$H = 2$	$H = 4$	$H = 8$	$H = 12$	$H = 16$	$H = 20$
$\hat{\beta}_{cay}^{\text{NDS}}$	0.241 (1.118) [0.001]	0.464 (1.092) [0.005]	0.811 (0.708) [0.009]	1.536 (0.722) [0.021]	2.134 (0.679) [0.032]	2.944 (0.996) [0.053]	3.451 (1.007) [0.058]
$\hat{\beta}_{cay}^{\text{PCE}}$	0.462* (1.698) [0.008]	0.977* (1.948) [0.020]	2.124 (1.541) [0.051]	4.408* (1.961) [0.122]	5.909** (2.290) [0.165]	7.101*** (3.762) [0.201]	7.741*** (3.327) [0.186]
$\hat{\beta}_{cay}^{\text{NDS10}}$	0.372* (1.723) [0.007]	0.722* (1.684) [0.014]	1.379 (1.237) [0.029]	2.689 (1.447) [0.062]	3.549 (1.527) [0.081]	4.643** (2.140) [0.117]	5.202* (1.790) [0.115]
$\hat{\beta}_{cay}^{\text{PCE10}}$	0.629** (2.417) [0.020]	1.268** (2.561) [0.041]	2.737** (2.227) [0.097]	5.607** (2.522) [0.224]	7.167*** (3.266) [0.274]	8.374*** (5.779) [0.318]	8.580*** (4.221) [0.264]

Note: *p<0.1; **p<0.05; ***p<0.01.

3.2. Predictive regressions

In their original work, Lettau and Ludvigson (2001) show that cay is a strong predictor for future excess stock returns even when controlling for several other common predictor variables. We focus on the predictability of H -period ahead excess returns using cay as the sole predictor, meaning we consider the (standard) predictive regression

$$\tilde{r}_{t,H} = \alpha_{cay}^k + \beta_{cay}^k \widehat{cay}_t^k + \epsilon_{t,H}^k, \quad k \in \{\text{NDS, PCE, NDS 10, PCE 10}\} \quad (12)$$

where $\tilde{r}_{t,H} = r_{t+1} - r_{f,t+1} + \dots + r_{t+H} - r_{f,t+H}$. In Table 3, we report the results for horizons of up to five years ($H = 20$).

In principle, (5) also allows for the possibility of cay to predict future consumption growth. In Appendix D, we present evidence that this is (still) not the case for the aggregate cay even though the predictability of asset returns has weakened over time.

The results displayed in Table 3 are striking. At the 1% significance level, we find none of the cay specifications to be significant predictors over the course of up to two years. The comparatively best predictor turns out to be $\widehat{cay}_t^{\text{PCE10}}$ for which we find significance at the 5% level for the shorter horizons. This is in line with our expectation for the top-10 specification as well as PCE being the

superior consumption measure. In case of NDS as the consumption measure, we also see a notable improvement of the top-10 specification against its aggregate counterpart, albeit on a lower level. It shows that the adjustment to the consumption and wealth of only the richest households significantly outperforms the aggregate specification, which is an indication that it may capture the underlying theory more adequately.⁹ We note that our results are derived from the methodology outlined above, which is to adjust aggregate series by wealth and income shares based on fitted values on the capital share, $K S_t$. As we discuss in [Appendix A](#), we obtain weaker, albeit qualitatively similar, results when using raw data. We further show that the projection on $K S_t$ introduces a meaningful economic adjustment that improves the stability of the cointegration relationship.

The adjusted R^2 statistics are well below one percent for a one quarter horizon when NDS is the consumption measure. For PCE they are only marginally better. These results are in contrast to findings in prior literature. For example, [Lettau and Ludvigson \(2001\)](#) report a coefficient estimate of 2.165 with a t -statistic of over three and an adjusted R^2 of 9% for a sample period from 1952:4–1998:3. Other papers (e.g., [Guo \(2006\)](#), [Hahn and Lee \(2006\)](#), [Della Corte et al. \(2010\)](#), [Sousa \(2010\)](#), [Guo et al. \(2013\)](#)) confirm these strongly significant findings for different sample periods. [Lettau and Ludvigson \(2001\)](#) also conclude that the impact of a one-standard deviation rise in $\widehat{cay}_t^{\text{NDS}}$ is economically large with expected excess returns going up by about 220 basis points quarterly, roughly a nine percent increase at an annual rate. With a current standard deviation of 0.024, we only get an (insignificant) increase of about 58 basis points in excess returns in the following quarter.

Taken together, the results lead us to conclude that the predictive ability of cay has fundamentally weakened over the last roughly two decades. We investigate this apparent decay more closely from various perspectives in what follows.

3.3. The decay in-sample

[Table 3](#) above documents the current weak predictive power of cay . We investigate next in more detail how the predictive ability has changed over time. We therefore recursively estimate the forecast regression (12) by adding one quarter of observations step by step. It takes a substantial amount of observations to reliably estimate the cointegration relation and since the cointegrating parameters do not seem to stabilize before at least 1990, we also do not begin our recursive estimation before that date. The results for all four of our specifications of cay are reported in [Fig. 3](#).

While the top plots of both panels directly show the coefficient estimate for β_{cay} from (12) with horizons of one and eight quarters, respectively, the bottom plots have more economic meaning. They represent the change in expected excess returns given a one standard deviation increase in $\widehat{cay}_t$ this quarter. We see that until the early 2000s the impact of cay was large. For NDS it was around 200bp whereas for PCE it was slightly lower but still well above 150bp. This changed dramatically until the end of the sample in 2019 when the impact of $\widehat{cay}_t^{\text{NDS}}$ fell below 60bp whereas the one of $\widehat{cay}_t^{\text{PCE}}$ fell to 90bp. This finding is in line with [Lettau and Ludvigson \(2019\)](#). It shows that the decay of cay is to some extent mitigated when switching to PCE instead of NDS. However, also for PCE we do find a significant decline in predictive power which cannot be explained by the falling ratio of NDS to PCE alone.

In any case, we see that our top-10 specifications perform considerably better than their aggregate counterparts during the last roughly 15 years. While trailing their counterparts slightly in the 1990s, they have since overtaken them and produce stronger forecasts. This effect seems to be even more pronounced after the global financial crisis of 2007/08 and the subsequent Great Recession. Nonetheless, they are not able to completely eradicate the decay of cay as for example $\widehat{cay}_t^{\text{NDS } 10}$ has also fallen below 90bp lately down from levels of around 200bp. Only $\widehat{cay}_t^{\text{PCE } 10}$ was able to produce a rather constant level of predictability before a mild decay set in after 2015. The impact on next quarter's excess return remains at a remarkably high 129bp though.

The results strongly confirm our belief in a reinforcement of the underlying consumption smoothing theory by focusing on the wealthiest households who are likely to better represent the marginal investor and are thus crucial when it comes to forming expectations on future returns.

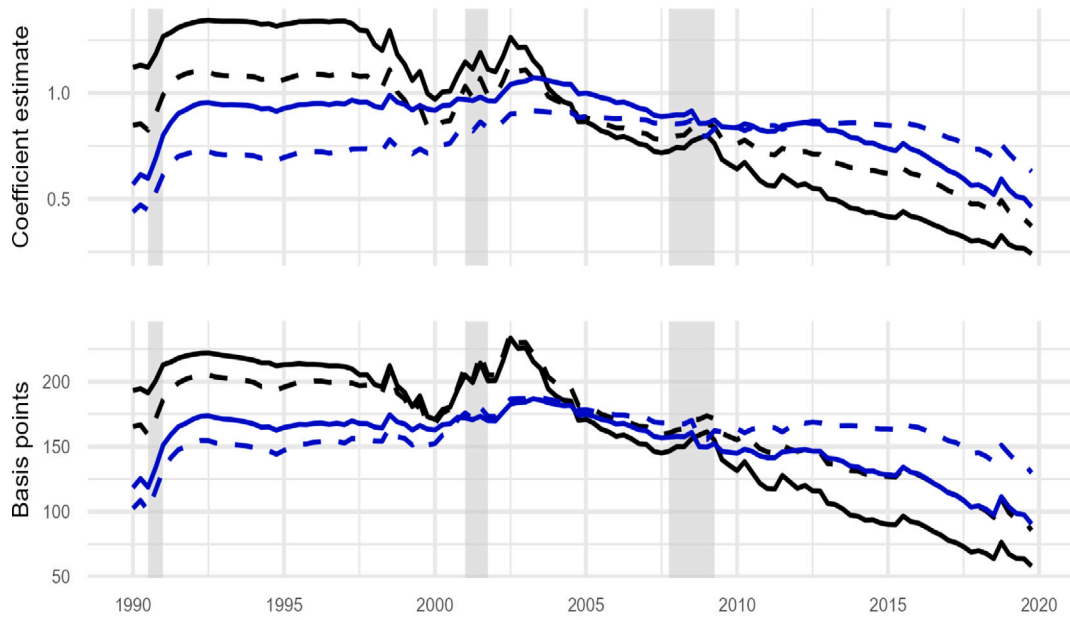
A similar picture emerges when considering the excess return over the next two years. In panel B of [Fig. 3](#), we see that the 1990s were again a very stable period for the coefficient estimate of all four specifications in which the aggregate versions of cay were stronger predictors than their top-10 counterparts. After peaking in the early 2000s, the coefficients of both NDS measures started declining whereas the PCE specifications fared much better. For the aggregate PCE specification a visible decline only sets in after 2015 while $\widehat{\beta}_t^{\text{PCE } 10}$ remains more or less stable until the end of the sample. The results for even longer forecast horizons look very much the same. This fact, and to avoid any potential overlapping issues in the forecasts, centers our focus on one-quarter ahead forecasts from here on out.

3.4. The decay out-of-sample

The estimation of cay naturally contains what [Lettau and Ludvigson \(2001\)](#) refer to as 'look-ahead' bias. The fact that the full sample is used to estimate the cointegrating parameters implies that cay contains future information. Therefore, forecast regressions build upon cay can potentially be spurious. [Lettau and Ludvigson \(2001\)](#) address this issue by comparing the mean squared error from forecasting regressions including a true out-of-sample cay that is periodwise re-estimated with the one from forecasting regressions that do not contain cay . Still, [Brennan and Xia \(2005\)](#) claim that cay does not perform better than a mechanistic variable called ι_{ay}

⁹ One concern regarding the significance of our proposed top-10 specification is the well-known [Stambaugh \(1999\)](#) bias. We refrain from reporting the adjusted coefficients to be consistent with the previous literature, but find that adjusting for the bias does not materially affect our results, as, e.g., the coefficient of $\widehat{\beta}_{cay}^{\text{PCE } 10}$ at the quarterly forecast horizon shrinks only slightly to 0.590, implying a marginally lower t -stat of 2.269.

Panel A: One-quarter ahead forecasts



Panel B: Two-year ahead forecasts

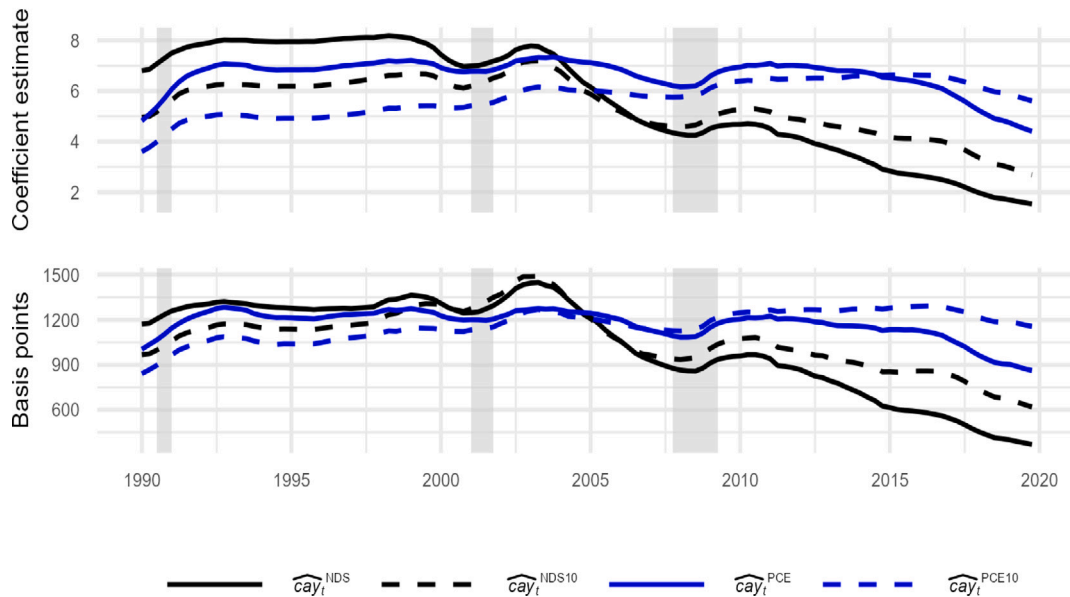


Fig. 3. Both panels show the quarterly re-estimated coefficient estimate $\hat{\beta}_{cay}$ for various versions of $\widehat{cay}_t$, in the forecast regression (12) of excess returns in the top plot. Panel A for one-quarter ahead and panel B for two-year ahead forecasts. The solid lines depict the aggregate specifications whereas the dashed lines depict the top-10 specifications. The results for NDS as the consumption measure are shown in black, the ones for PCE in blue. The sample period is 1952:1–2019:4. The gray vertical bars mark NBER recessions. Likewise, the bottom plots in each panel show the increase of excess returns in basis points given a one standard deviation increase in the respective $\widehat{cay}_t$.

that simply replaces consumption with a linear time trend in the cointegration regression. In a direct reply to this critique, Lettau and Ludvigson (2005b) point out that for a correct estimation of the cointegrating parameters it is inevitable to use the full sample. They argue that the parameters in cay are steady state wealth shares that are perfectly known to a representative agent in equilibrium. Furthermore, they claim that the deterministic time trend in lay contains more economic content than one would naively expect and

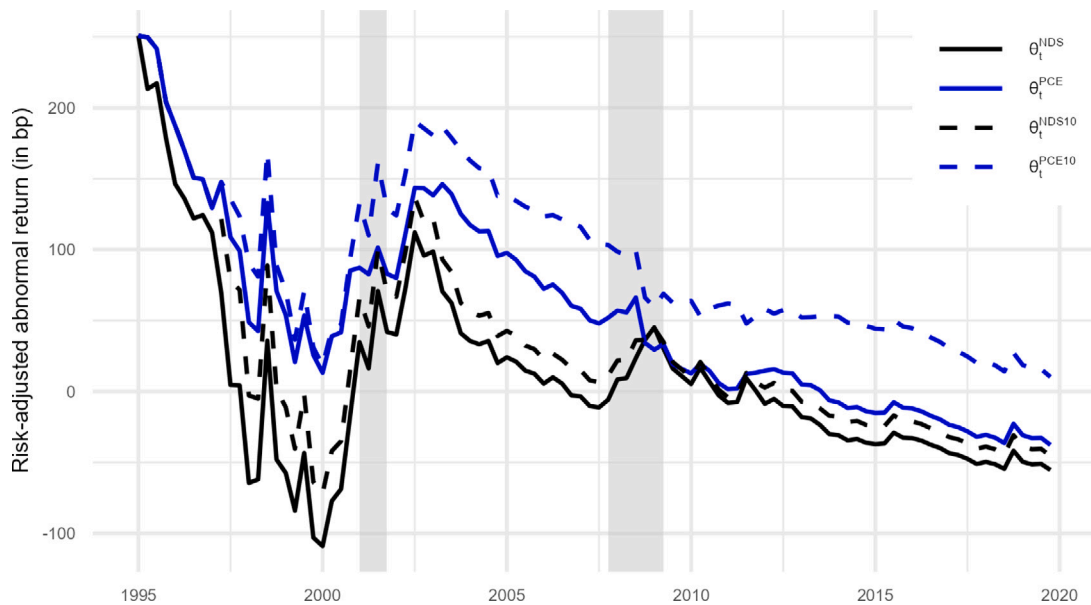


Fig. 4. Risk-adjusted abnormal return θ_t of all four *cay*-strategies with respect to the historical mean strategy in basis points. The sample period is 1952:1–2019:4. The cointegration parameters are largely stable around 1995, see Fig. 5, which is why the out-of-sample forecasts begin in the first quarter of that year.

that it is perfectly consistent with forward looking behavior and optimization. In another paper, [Hahn and Lee \(2006\)](#) also address the issue of the ‘look-ahead’ bias in *cay*. We will come back to this issue in more detail in Section 4.

The most notable contribution on the out-of-sample performance of *cay* has been the seminal work of [Welch and Goyal \(2008\)](#). They outline an abysmal out-of-sample forecasting power of *cay* next to several other common predictor variables. Since the poor (proper) OOS performance of *cay* has been already well documented in the literature, we refrain from repeating this exercise in this paper, but refer to [Appendix E](#), where corresponding results are reported.

Instead of reporting (pure) OOS results, we analyze the economic gain from an investor’s perspective that can (or rather could) be harvested from implementing an investment strategy based on forecasts using *cay* against a ‘naive’ rolling historical mean strategy.¹⁰ This approach closely follows [Della Corte et al. \(2010\)](#) in assuming a (short-selling constrained) mean–variance investor, who allocates wealth between a risk-free asset and the market portfolio. The investor can quarterly re-allocate wealth with the optimal weight on the market in period t given by

$$w_t = \frac{1}{\lambda} \frac{E_t [r_{t+1} - r_{f,t+1}]}{\text{Var}_t [r_{t+1} - r_{f,t+1}]} \quad (13)$$

where λ is the coefficient of relative risk aversion (RRA) and E_t and Var_t denote conditional expectation and variance, respectively. We follow [Campbell and Thompson \(2008\)](#) in assuming λ to be three.

The ex-post performance of the *cay*-strategy relative to the rolling historical mean strategy is evaluated by means of risk-adjusted abnormal returns from [Goetzmann et al. \(2007\)](#).¹¹

Our sample contains 68 years of quarterly observations which we split in 1995 to use the first part as a training period to then evaluate the out-of-sample predictability in the second part of the sample. This relatively long training period is reasonable given that it takes time until the cointegrating parameters reach their steady state values, see Fig. 5 below. The risk-adjusted abnormal return θ measures the outperformance of the *cay*-strategy with respect to the benchmark historical mean strategy. The development of θ over time is shown in Fig. 4.

¹⁰ To be precise: For a historically accurate out-of-sample strategy back-test, we would need to employ real-time consumption data as the consumption series are constantly revised by the BEA. Since the estimation of $\widehat{cay}_t$ does not only require such real-time consumption series but also various other macroeconomic series such as the asset wealth and labor share series as well as all the series that enter the construction of the labor income series, it is effectively infeasible to perform such an exercise. We therefore chose to assess the *cay*-strategy by using the current versions of all series published by the BEA (and BLS), being well aware that this procedure still contains future information flowing into the revision process of the data. However, we believe it still contains economic content on whether a constantly evolving strategy based on forecasts employing $\widehat{cay}_t$ as a predictor could have outperformed a basic mean strategy over the course of our sample. Specifically, it only utilizes data theoretically available at a certain point in time albeit this data practically not being observable back then. In this sense our analysis is out-of-sample with respect to the true data but not the temporarily available data. We believe this to be perfectly in line with the underlying equilibrium model for the same reasons that [Lettau and Ludvigson \(2005b\)](#) indicate.

¹¹ We refrain from reporting a detailed description of the methodology here and refer to the exhaustive description in [Della Corte et al. \(2010\)](#) instead.

We see that at the beginning of our sample the aggregate and top-10 specifications performed rather similar and outperformed the historical average strategy by roughly 200bp. Beginning around the end of the 1990s the top-10 versions begin to outperform their aggregate counterparts, a finding that holds true until the end of our sample. After a short resurgence in the early 2000s to levels of above 100bp, the, by now familiar, decay of cay sets in until all specifications, except $\widehat{cay}_t^{PCE10}$, exhibit a double-digit negative θ at the end of our sample. The roughly 10 basis points left for the top-10 specification using PCE is again consistent with our belief that this alteration of cay captures the underlying theory best, although any sizable advantage over a historical mean strategy seems also practically gone.

4. A structural shift in the cointegration relation

Having documented the substantial decay of cay in the previous section, we now shed some light on the origin of this declining forecasting power. Recall again, that cay is expected to be a viable predictor for stock market returns since it is derived as the residual from the (single) cointegration equation between consumption, aggregate wealth and labor income and therefore proxies for the consumption-wealth ratio. The assumption of a stable equilibrium relationship may be justified from the rationale of economic theory as invoked by Lettau and Ludvigson (2001), who argue that the parameters in cay are steady state wealth shares which are perfectly known to an agent in equilibrium. Our previous section on results followed this reasoning in the sense that we employed cay without questioning its properties as a stationary cointegrating residual. However, the existence of a stable equilibrium relationship is ultimately an empirical question, which we investigate in more detail in this section. Testing cointegration relations requires a number of modeling choices on the part of the econometrician, such as the existence of trends, the number of lags and allowance of intercepts. We first conduct different cointegration tests and analyze the results of a Vector Error Correction Model.

4.1. Stability of the cointegrating parameters

In their reply to Brennan and Xia's (2005) criticism of a 'look-ahead bias', Lettau and Ludvigson (2005b) emphasize the importance of using the full available sample to estimate the cointegrating parameters. Taking Lettau and Ludvigson (2005b) at face value leads to the cointegrating parameters already reported in Table 1. In panel A of Table 4, we contrast these results to the cointegrating parameters from several other studies that used a different sample period. It is apparent that the coefficient of asset wealth has decreased and at least partly lost its significance, while the coefficient of labor income has increased towards one. This already speaks against a stable long-run equilibrium relationship between the three series forming cay as otherwise we would not expect those coefficients to vary much over time.¹²

To elaborate further on this preliminary finding, we quarterly re-estimate the cointegration relationship in an expanding window beginning in 1990 and show the results for $\hat{\beta}_{a,t}$ and $\hat{\beta}_{y,t}$ in Fig. 5.

The top left plot shows stable values for $\hat{\beta}_{a,t}^{NDS}$ for most of the 1990s, from which point on the coefficient of asset wealth steadily declined. Since the cointegrating parameters should in theory add up to one, it is no surprise that $\hat{\beta}_{y,t}^{NDS}$ exhibits the mirrored behavior with an increase towards values close to one. This picture is essentially the econometric confirmation of what can be observed from the development of the underlying data series that cay is build upon as illustrated in the upper panel of Fig. 1. After detrending the consumption, asset wealth and labor income series, we can observe that, beginning with the late 1990s, asset wealth behaves increasingly detached from its (alleged) long-run relationship with consumption and labor income. It is reasonable to assume, that this detachment is related to the increasing wealth inequalities, particularly after the Global Financial Crisis. Therefore, we are not surprised to see our cointegration estimates for our proposed top-10 specification using PCE being stable after the initialization period as can be seen in the bottom right plot of Fig. 5. Thus, the existence of a stable long-run equilibrium relationship seems to be justified, especially in comparison to the aggregate NDS specification.

Ultimately, we want to address the question whether there even exists a valid cointegration relation between these series (anymore). To answer this question we report the test statistics of residual based Phillips and Ouliaris (1990) cointegration tests in panel B of Table 4. We do so not only for our full sample but also for the original sample period in Lettau and Ludvigson (2001). Similar results are obtained when employing Engle and Granger (1987) or Johansen (1988, 1991) tests as in Appendix C.2.¹³

Within the full sample we are far from rejecting the null of no cointegration. The closest we get is in the case of PCE10 with a p -value of roughly 22%. Interestingly, we also cannot reject the null in the Lettau and Ludvigson (2001) sample period until the third quarter of 1998. We are thus not able to replicate their test results.¹⁴ Despite this failure to replicate the exact results in Lettau and Ludvigson (2001), panel B of Table 4 again confirms our finding that PCE10 holds up reasonably well in comparison to the other three specifications.

Finally, to assess the short-term dynamics of the cointegration relationship, we report the results of a Vector Error Correction Model (VECM) for both sample periods in Table 5. We restrict ourselves to only showing the results for NDS and PCE10 here as

¹² Note, that by restricting our data to the same samples as in these papers, we are able to replicate the results qualitatively up to small deviations due to data revisions.

¹³ As pointed out in the introduction to this section, cointegration tests are subject to different modeling choices. We stick to the specification, where we allow a linear trend in the data (but none in the cointegration relation) and an intercept in the cointegration relation. We discuss the particular role of the deterministic time trend in the cointegration equation in more detail in the later Section 4.2.

¹⁴ Our findings are in line with Rudd and Whelan (2006). It is also interesting to observe, that in their latest assessment of cay , Lettau and Ludvigson (2019) prefer to report results from the Ogaki and Park (1997) test, which reverses the null and ultimately does not allow them to reject the null of cointegration.

Table 4

Panel A shows the cointegration parameters of our four specifications next to the estimates of several other studies. Thereby LL2001 denotes [Lettau and Ludvigson \(2001\)](#), HL2006 denotes [Hahn and Lee \(2006\)](#), DC2010 denotes [Della Corte et al. \(2010\)](#) and LL2019 denotes [Lettau and Ludvigson \(2019\)](#). Panel B shows the test statistics from [Phillips and Ouliaris \(1990\)](#) cointegration tests of our four specifications over different sample periods.

Panel A: Cointegration parameters								
	aggregate NDS				aggregate PCE		top-10 NDS/PCE	
	NDS 52:1–19:4	LL2001 52:4–98:3	HL2006 52:4–02:4	DC2010 1946–2006	PCE 52:1–19:4	LL2019 52:1–14:3	NDS10 52:1–19:4	PCE10 52:1–19:4
$\hat{\beta}_a$	0.029 (0.809)	0.310*** (11.700)	0.275*** (27.500)	0.274*** (11.417)	0.235*** (8.148)	0.304 NR	0.091** (2.081)	0.280*** (9.981)
$\hat{\beta}_y$	0.929*** (24.243)	0.590*** (23.920)	0.616*** (61.600)	0.684*** (28.500)	0.786*** (25.072)	0.729 NR	0.874*** (19.306)	0.766*** (27.810)

Note: *p<0.1; **p<0.05; ***p<0.01.

Panel B: Cointegration tests						
specification	90% CV	95%CV	1952:4–1998:3		1952:1–2019:4	
			Statistic	p-value	Statistic	p-value
NDS	–3.449	–3.768	–2.732	0.369	–1.621	0.868
PCE	–3.449	–3.768	–3.309	0.144	–2.327	0.577
NDS10	–3.449	–3.768	–2.757	0.357	–1.734	0.834
PCE10	–3.449	–3.768	–3.277	0.153	–3.082	0.215

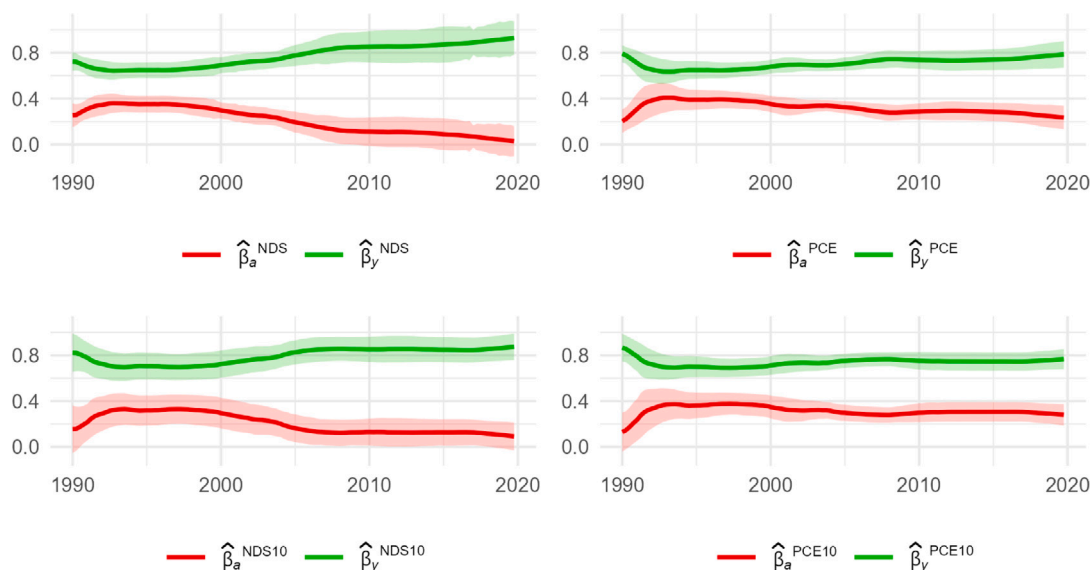


Fig. 5. Cointegrating parameters for asset wealth (red) and labor income (green) quarterly re-estimated. The shaded areas represent the corresponding confidence bands of ± 2 standard errors. The sample period is 1952:1–2019:4.

these two are the most interesting specifications from what we have seen so far. The most important coefficient is clearly the one of $\widehat{cay}_{t-1}$ in the regression of Δa_t (columns 2 & 5). Similar to [Lettau and Ludvigson \(2001\)](#), we find it to be positively significant for NDS with a value of 0.175 in the sample until 1998:3. This shows that transitory movements in asset wealth are rather foreshadowed by cay than by consumption or labor income growth. However, $\widehat{cay}_{t-1}$ is far from significant (t -stat of 0.499) from the perspective of most recent data. Consistent with the results from the cointegrating regression, this leads to the conclusion that cay does not (or rather, no longer) predict subsequent (transitory) changes in asset wealth. For the top-10 specification using PCE we also see a decline in the coefficient of $\widehat{cay}_{t-1}$ that goes along with a reduction in the t -stat but we still have significance at the 5% level, once more confirming that adjusting the consumption–wealth ratio for the richest households realigns the empirics with the theory.

This result raises two issues. First, the sustained shift in parameters opens the question of allowing for a deterministic time trend in the cointegration relationship. We will address this separately in the following subsection. Second, from the pure perspective of the econometrician (and in line with the argumentation of [Lettau and Ludvigson, 2005b](#)), parameter estimates from the most comprehensive data sample should be used. From an economic theory standpoint, however, one may argue that some equilibrium relationship has to exist and maybe the [Lettau and Ludvigson \(2001\)](#) parameters do best represent it. We therefore ask, how holding

Table 5

Vector Error Correction Models (VECM) for NDS and PCE10 with two different sample periods each.

Panel A: Vector Error Correction Model for NDS						
	1952:4-1998:3			1952:1-2019:4		
	Δc_t	Δa_t	Δy_t	Δc_t	Δa_t	Δy_t
Δc_{t-1}	0.206** (2.485)	0.160 (0.582)	0.468*** (3.014)	0.339*** (5.448)	0.228 (0.856)	0.619*** (4.669)
Δa_{t-1}	0.065*** (2.770)	0.198** (2.544)	0.087** (1.976)	0.054*** (3.642)	0.150** (2.383)	0.049 (1.563)
Δy_{t-1}	0.067 (1.403)	-0.114 (-0.721)	0.001 (0.007)	0.039 (1.248)	-0.110 (-0.821)	-0.124* (-1.863)
$\widehat{cay}_{t-1}$	-0.021 (-1.016)	0.175** (2.578)	0.007 (0.193)	0.003 (0.264)	0.021 (0.499)	0.032 (1.499)
$\bar{R}^2$	0.162	0.051	0.090	0.221	0.015	0.099
Note: *p<0.1; **p<0.05; ***p<0.01.						
Panel B: Vector Error Correction Model for PCE10						
	1952:4-1998:3			1952:1-2019:4		
	Δc_t	Δa_t	Δy_t	Δc_t	Δa_t	Δy_t
Δc_{t-1}	0.206** (2.466)	0.070 (0.256)	0.459*** (2.939)	0.345*** (5.536)	0.172 (0.650)	0.624*** (4.700)
Δa_{t-1}	0.066*** (2.834)	0.191** (2.494)	0.087** (1.982)	0.053*** (3.618)	0.159** (2.533)	0.052 (1.647)
Δy_{t-1}	0.077* (1.671)	-0.158 (-1.043)	0.001 (0.016)	0.035 (1.124)	-0.073 (-0.547)	-0.130* (-1.950)
$\widehat{cay}_{t-1}$	-0.005 (-0.342)	0.170*** (3.376)	0.014 (0.494)	-0.006 (-0.487)	0.112** (2.241)	0.027 (1.058)
$\bar{R}^2$	0.158	0.075	0.091	0.222	0.033	0.095
Note: *p<0.1; **p<0.05; ***p<0.01.						

fixed a set of parameters influences the predictive ability of cay over time. This means, contrary to our previous analysis in Section 3, that we now fix the cointegrating parameters instead of re-estimating them sequentially over an expanding window. We use two parameter sets: The most recent one (i.e. parameters $\hat{\beta}_a$ and $\hat{\beta}_y$ from Table 1), and the one for the Lettau and Ludvigson (2001) sample period from 1952:4–1998:3 (i.e. $\hat{\beta}_a = 0.328$ and $\hat{\beta}_y = 0.668$).¹⁵ We plot the coefficients from the predictive regression of (12) for both cases in Fig. 6. We restrict ourselves to the classical (aggregate) NDS specification here since it is the one that experiences the shift in the cointegration relation most notably.

As known from Lettau and Ludvigson (2001), their parameter set worked well for (in-sample) prediction until 2000, leading to an almost perfectly stable forecasting ability. However, the panel B of Fig. 6 also shows that even by arguing that this parameter set is the “correct” one (and therefore holding it fixed), we find the same decay in forecasting power as with the re-estimated parameters in Fig. 3. Vice versa, by using the most recent parameter set, the upper panel indeed shows us that cay would have never worked well with the predictive coefficient being on the verge of significance at the 10% level constantly.

4.2. The role of a deterministic time trend

In this subsection, we assess the possibility that the structural shift in the cointegrating parameters could be caused by an omitted deterministic time trend. This is ultimately a question of model choice regarding the cointegration relationship, an issue extensively discussed in Koop et al. (2008). The relevance of an omitted time trend in the cointegration relationship was first brought to attention by Brennan and Xia (2005), who argue that a mechanistic variable, which they call tay , that simply replaces consumption with calendar time in the estimation of the cointegration relationship, predicts asset returns at least as good as cay .¹⁶ A more detailed analysis of a possible time trend is provided by Hahn and Lee (2006), who show that the restricted version of cay (i.e., without time trend) from Lettau and Ludvigson (2001) is just the sum of a bias component and the unrestricted cay estimated with a deterministic

¹⁵ We estimate the coefficients for the LL2001 sample period on today's data to be consistent with our estimation that uses the full sample. The difference to the reported coefficients in Lettau and Ludvigson (2001) is a consequence of data revisions as we are able to successfully replicate the estimations in Lettau and Ludvigson (2019) that also uses more recent data.

¹⁶ In their strong reply to Brennan and Xia (2005), Lettau and Ludvigson (2005b) not only argue that from a theoretical standpoint, a meaningful long-run equilibrium between consumption, aggregate wealth and labor income does not allow for a deterministic trend. They also claim that tay contains more economic content than acknowledged by Brennan and Xia (2005), by the mere fact that tay is nothing but a proxy for cay due to a large part of the variability of consumption being governed by a simple deterministic time trend. A forward-looking consumer anticipates this relationship which brings it perfectly in line with the model assumptions.

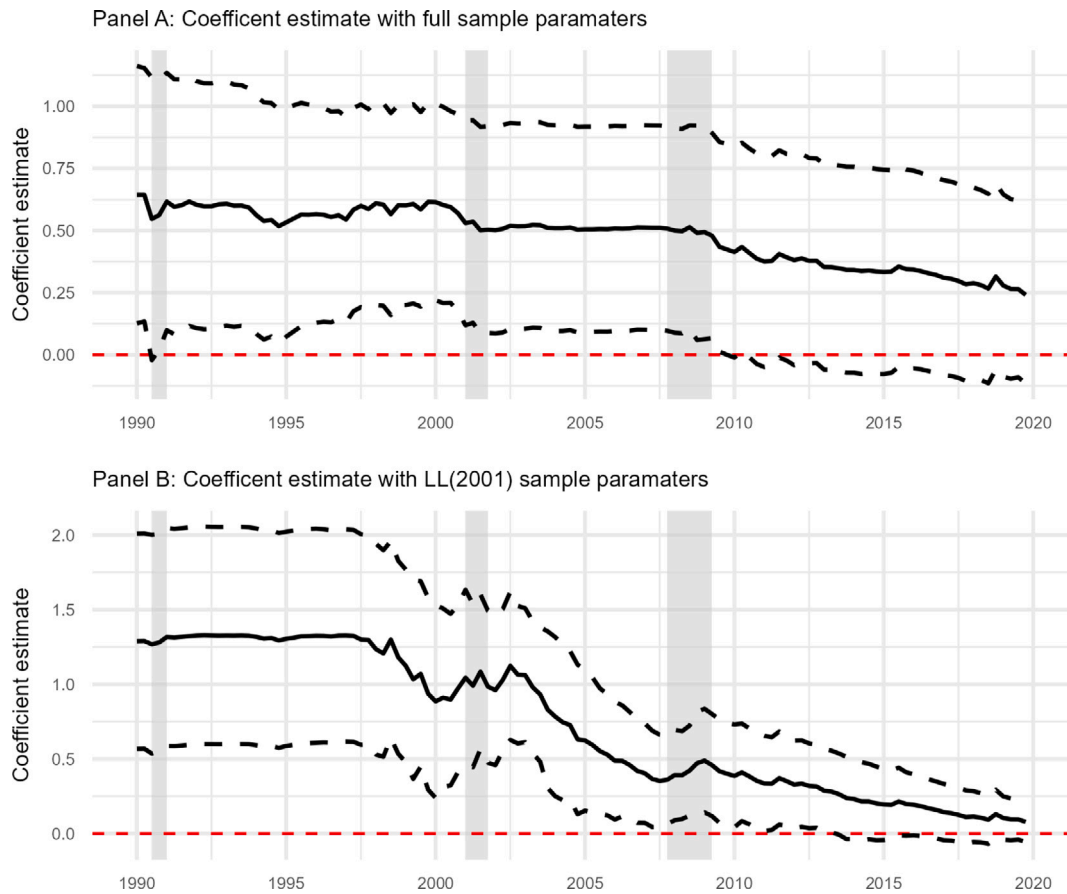


Fig. 6. Both plots show the period wise re-estimated coefficient estimate of $\widehat{cay}_t$ in quarterly forecasts of excess returns accompanied by its 90% confidence band based on Newey and West (1987)-standard errors as the dashed lines. The top panel represents the results using the cointegrating parameters estimated using the full sample from 1952:1–2019:4, while the bottom panel uses the cointegrating parameters from the sample 1952:4–1998:3.

time trend. Their main finding is that the time trend takes away a substantial part of the forecasting ability of *cay*, or in other terms, that the unrestricted *cay* (with trend) does show substantially less forecasting power.

To investigate the role of the time trend, we follow Hahn and Lee (2006) and augment Eq. (10) by calendar time t and derive *cay* from $\widehat{cay}_t = c_t - \hat{\pi}t - \hat{\beta}_a a_t - \hat{\beta}_y y_t$, where $\hat{\pi}$ is the estimated coefficient on time t . We estimate the cointegration regression sequentially in an expanding window and report the time variation of the coefficients $\hat{\pi}_t$, $\hat{\beta}_{a,t}$, and $\hat{\beta}_{y,t}$ in Fig. 7.

We find that the deterministic time trend was strongly significant until around 2010 (with t -statistics of more than 6) for both aggregate specifications, but sharply dropped thereafter to become insignificant from the perspective of most recent data. For the two top-10 specifications the time trend is persistent and even increased slightly over the sample period. The inclusion of the time trend has the remarkable effect that the coefficient on wealth, $\hat{\beta}_{a,t}$, would have never been significantly different from zero when NDS is the consumption measure, which in turn leads to the result that *cay*, estimated from the unrestricted model, would have never displayed forecasting power. This result reinforces the findings of Hahn and Lee (2006) and their conclusion that the good (in-sample) forecasting performance of *cay* was (at least partly) due to omitting a time trend. However, note that the situation for the aggregate specifications changes dramatically after 2010, where the inclusion of the time trend is no longer warranted from an econometric point of view. Furthermore, note that the decline of the time trend goes along with a substantial increase in $\hat{\beta}_{y,t}$, i.e., the coefficient on income after 2010. So, the time trend apparently absorbs the impact of wealth prior to 2010, but we still observe a similar structural shift as already documented above in Fig. 5. For the top-10 specifications, especially the one using PCE, all three coefficients are more or less stable since the early 1990s and there is no comparable shift as in the upper two panels. In Appendix A, we show the same plot for the specification where the raw data on wealth and income shares is used, and which shows that the coefficient instability in this case is even worse than in the case of the aggregate *cay*.

In sum, the analysis of a time trend reveals some irony of history. Prior to 2010, the original *cay* showed good (in-sample) forecasting power, but was subject to the potential criticism that it earned its predictability due to the omitted (but statistically significant) time trend. Since 2010, the data does support the theoretically motivated assumption of omitting a time trend, but

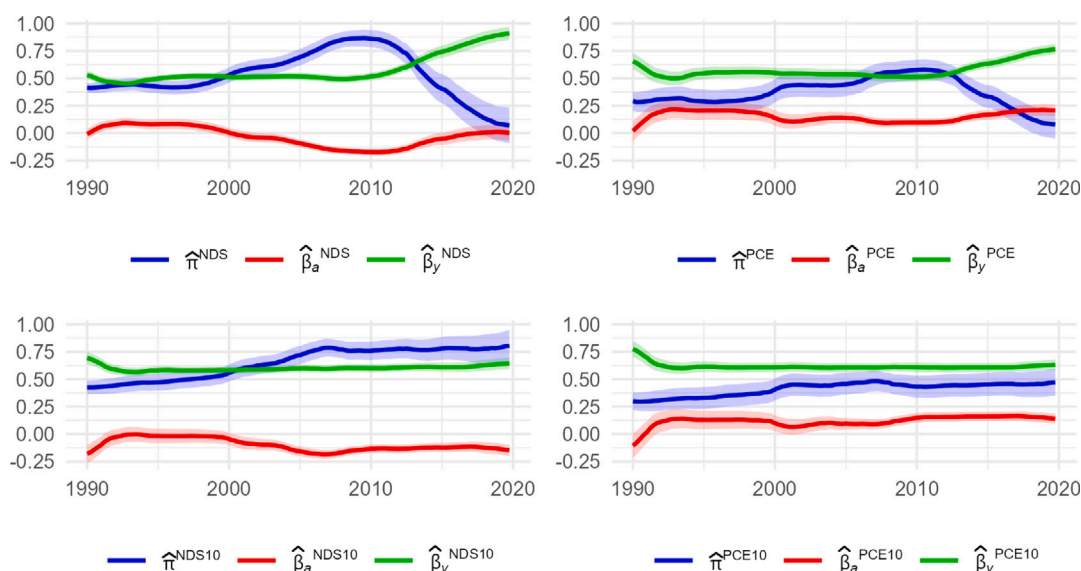


Fig. 7. Cointegrating parameters for asset wealth (red), labor income (green) and a time trend (blue) quarterly re-estimated. The shaded areas represent the corresponding confidence bands of ± 2 standard errors. The sample period is 1952:1–2019:4.

even the restricted *cay* no longer displays forecasting power. Our *cay* for the top 10% richest households, however, exhibits again a more consistent behavior through time, although the critique of an omitted time trend in the cointegration relation might also be warranted in this case.

5. Alternative improvements of *cay*

Since the inception of *cay* in Lettau and Ludvigson (2001), a number of alternative constructions have been proposed in the literature. In this section, we will elaborate on three of them in more detail that we believe are capturing different aspects in terms of estimating the forecasting ability of the consumption–wealth ratio.

The attempts to enhance the forecasting ability of the (estimated) consumption–wealth ratio can broadly be classified into two strands of literature. The first and larger one aims at improving the first step of the estimation, namely developing a better proxy for the unobservable consumption–wealth ratio. Several procedures have been proposed, from a present value approach by Ren et al. (2014), over an FVAR-approach by Quineche (2021) to a Markov-switching version of the consumption–wealth ratio by Bianchi et al. (2022). The one that comes conceptually closest to our top-10 specification is *cday* from Sousa (2010). The main innovation there is to disaggregate wealth, meaning to split up (aggregate) asset wealth into (aggregate) financial wealth and (aggregate) housing wealth. In doing so, Sousa (2010) also questions the suitability of *aggregate* asset wealth as the optimal variable for the estimation of *cay*. Contrary to our top-10 specification, however, Sousa (2010) does this not from the standpoint of wealth inequalities but instead because he believes transitory movements in asset wealth are rather to be linked to transitory movements in the financial wealth component than in the housing wealth component. In a sample ending with the last quarter of 2008 he shows a stronger performance of *cday* compared to *cay*. Employing *cday* instead of *cay* constitutes our first alternative.

The second strand of literature consists of attempts which add further variables to the actual forecasting regression to address a potential omitted variables issue. Our second alternative goes in this direction and follows (Guo, 2006) by adding aggregate stock market volatility as a second regressor in the predictive regressions. Guo (2006) shows that this not only improves the significance of the coefficient estimate of *cay* but also the share of explained variation in terms of R^2 statistics.¹⁷

Our third alternative originates from the more general consumption-based asset pricing literature, picking up on an issue raised by Savov (2011) that the published consumption series from the BEA are subsequently smoothed (or filtered) and thus may lose important information from an asset pricing perspective. In a very similar spirit, Kroencke (2017) develops a model that imitates the filtering procedure undertaken by the BEA. He ultimately inverts this procedure to gain a proxy for the (more volatile) unfiltered original consumption series which is only observed by the BEA. Our third alternative applies said procedure to unfilter our consumption series before we use them in the cointegration relation from which we derive $\widehat{cay}$. All details on the three alternatives considered in this section can be found in Appendix F.

Despite the very different motivations underlying the three alternatives described above, they all share the potential to improve the forecasting ability of the consumption–wealth ratio. It thus seems natural to let them compete in a horse race against our

¹⁷ Similarly strong results are found in the later work of Guo et al. (2013).

Table 6

Coefficient estimates of the respective proxy for the consumption–wealth ratio from the one-quarter ahead predictive regression of excess returns. Newey and West (1987)-adjusted t -statistics appear in parentheses. Bold estimates indicate significance at the 5%-level. Adjusted R^2 -statistics are reported in square brackets. The four left most columns refer to the sample period until 1998:3, while the four right most columns show the results for the whole sample ending in 2019:4. Model (1) is our top-10 specification cay^{PCE10} . (2) is $cday$ from Sousa (2010). (3) represents the addition of volatility in the forecast regression as in Guo (2006), labeled cay^σ . (4) is the standard cay -measure using the unfiltered consumption series due to Kroencke (2017), labeled cay^{unfil} . All models use personal consumption expenditures (PCE).

1952:1–1998:3				1952:1–2019:4			
(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
cay^{PCE10}	$cday$	cay^σ	cay^{unfil}	cay^{PCE10}	$cday$	cay^σ	cay^{unfil}
0.780	1.273	0.976	0.564	0.629	0.497	0.458	0.316
(2.585)	(2.678)	(2.781)	(2.006)	(2.417)	(1.693)	(1.732)	(1.299)
[0.036]	[0.041]	[0.052]	[0.015]	[0.020]	[0.009]	[0.004]	[0.003]

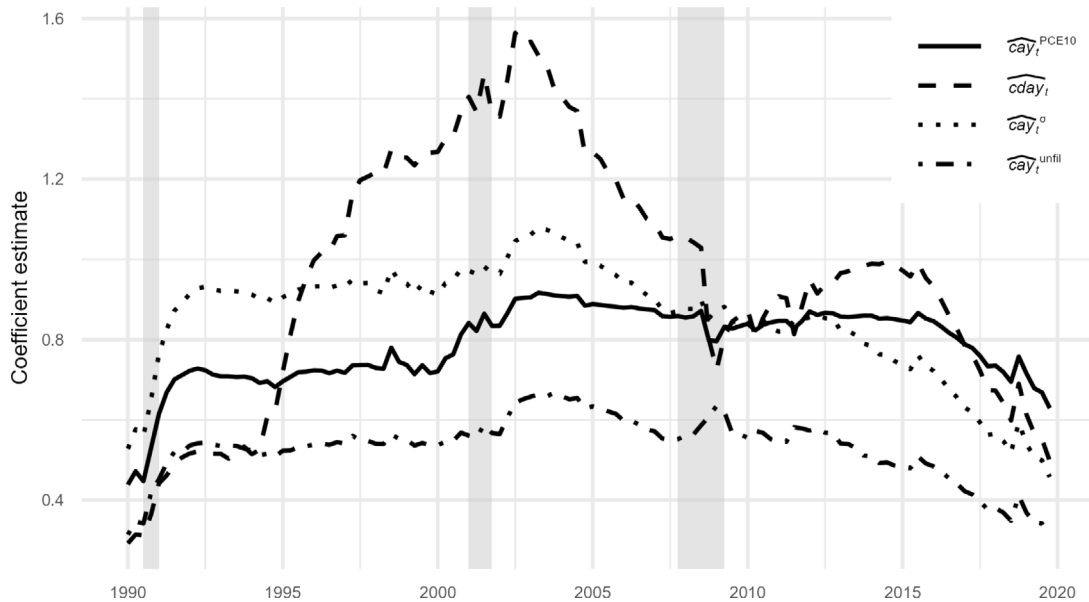


Fig. 8. The plot shows the quarterly re-estimated coefficient estimates of $\widehat{cay}_t^{PCE10}$, $\widehat{cday}_t$, $\widehat{cay}_t^\sigma$ and $\widehat{cay}_t^{unfil}$ in the forecast regression (12) of one-quarter-ahead excess returns. The sample period is 1952:1–2019:4.

proposed version of the consumption–wealth ratio for the rich. We do so by comparing the predictive regressions of quarterly excess returns of all four models in terms of significance of the respective cay -coefficients and adjusted R^2 -statistics. To see changes over the course of the last two decades where the decay of cay has manifested, we compare estimates for the samples ending in 1998:3 (corresponding to the end of the original Lettau and Ludvigson (2001) sample) and 2019:4. In all cases we only consider PCE as the consumption measure since it has proven to yield stronger results than NDS. The results of this exercise can be seen in Table 6.

While in the sample until 1998:3 all four models deliver significant coefficients with t -stats (well) above two, this has dramatically changed over the course of the subsequent two decades. The only estimate that still produces a significant coefficient estimate in the quarterly ahead forecasts is our proposed top-10 specification. Although we see a slight decrease both in the estimate itself as well as its t -statistic, it is still way above the threshold for significance at the 5%-level. In comparison, the other three alternatives are all well below this threshold and seem to be closer to the 10% significance level. The adjusted R^2 -statistics paint a very similar picture, with our proposed version clearly outperforming the alternatives in the full sample.

To provide more evidence on how the different cay versions evolved over time, we show the development of the predictive regression coefficients since 1990 in Fig. 8. It can be clearly seen that our suggested cay^{PCE10} exhibits the most stable behavior over time, and in particular is the only version among the alternatives considered which is significant by the end of our sample period. This, once again, indicates that our consideration of the richest households when measuring the consumption–wealth ratio mitigates the decay of cay much better than many other attempts which were shown to improve the original cay in past contributions.

We conclude that among the different avenues that try to make the theory work empirically, the identification of the representative agent in the presence of wealth inequalities appears to be a promising way forward. A full-fledged theory of cay under such inequalities is, however, beyond the scope of our contribution in this paper.

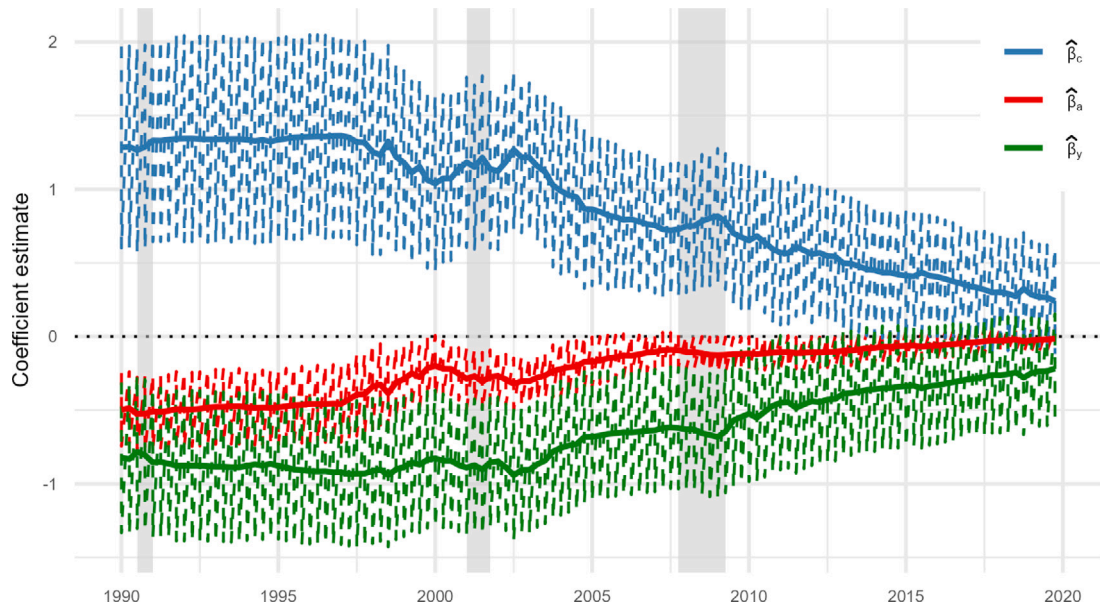


Fig. 9. The plot shows the coefficient estimates of NDS consumption (blue), aggregate wealth (red) and labor income (green) accompanied by their respective 90%-confidence bands based on Newey and West (1987)-corrected standard errors. The gray shaded areas mark NBER recessions. The sample period is 1952:1–2019:4.

6. Robustness

We conduct several robustness checks. As discussed already in the previous sections, we check different cointegration tests (see Appendix C) and conduct proper out-of-sample tests (see Appendix E).

Additionally, we check if *cay* also possesses predictive ability for consumption growth. Consistent with Lettau and Ludvigson (2001), we find no predictability for the aggregate version of *cay* using either NDS or PCE. The same holds for the top-10 specification using NDS. Only when looking at cay^{PCE10} , we find predictability at the one-quarter horizon and again at horizons of three or more years, see Table D.1. It is important to note, however, that this cannot be seen as evidence against the significant predictability in the predictive regression of returns in a Cochrane (2008)-style argument. The reason for that is the fact that the power of Cochrane's (2008) joint test relies on a “feature of the data” in which shocks to returns in the predictive regression are strongly negatively correlated with shocks to the dividend yield. This negative correlation is much less pronounced in the context of the consumption–wealth ratio, effectively taking away any power from the Cochrane (2008) test.¹⁸ Therefore, the significance of the coefficient estimate has to be judged on its own. This perfectly aligns with our finding that the (Stambaugh, 1999) bias does not materially affect our results due the exact same reasons.

Next, we address the issue of circumventing the construction of the cointegrating residual altogether, as discussed in Lettau and Ludvigson (2005a). Confronted with the criticism that the construction of *cay* involves the alleged ‘look-ahead’ bias, they argue that one can avoid the two-step approach and directly estimate a multivariate regression of excess returns on consumption, aggregate wealth and labor income. Therefore, we investigate whether the decay of *cay*, i.e., the disappearance of the forecasting ability of *cay* can also be found in the direct multivariate predictive regression. In line with Lettau and Ludvigson (2005a), we estimate

$$\tilde{r}_{t,H} = \alpha + \beta_c c_t + \beta_a a_t + \beta_y y_t + \varepsilon_{t,H} \quad (14)$$

for all four of our specifications. The results for horizons up to five years using the full sample can be found in Appendix G.

Fig. 9 illustrates the development of the quarterly re-estimated coefficients using aggregate NDS consumption over time. The previously documented decay of *cay* also manifests there in the sense that all three coefficient estimates have lost their significance and ‘decayed’ towards zero. The fact that the coefficient on consumption $\hat{\beta}_c$ (blue line) shows an almost identical pattern as the coefficient on $\widehat{cay}_t$ itself from Fig. 3 further strengthens our findings. This result is in line with asymptotic theory which implies that both approaches converge to the same estimate in large samples.

Finally, we provide results for updating our series to include the Covid-19 episode in Appendix H. None of the checks do materially affect our main conclusions.

¹⁸ Cochrane (2008) reports this correlation to be -70%. Other samples show correlations up to -90%. The correlation in our data is only -31%.

Table A.1

Panel A reports the coefficient estimates with standard OLS t -statistics in parentheses and adjusted R^2 statistics of the income and wealth shares regressed on the capital share. Panel B shows the coefficient estimates with Newey and West (1987)-adjusted t -statistics in parentheses and adjusted R^2 statistics in square brackets for the predictive regressions of one-quarter ahead excess returns on cay^{top10} for both fitted and raw series. The sample period is 1952:1–2019:4.

Panel A. Regression on capital share				
$X \in \{A, Y\}$	AS^{top10}		YS^{top10}	
	WID	SZZ	WID	
$\hat{\alpha}_X$	0.530*** (14.734)	0.498*** (8.746)	0.012 (0.731)	
$\hat{\beta}_X$	0.451*** (4.076)	0.465*** (2.687)	1.147*** (23.700)	
R^2	0.189	0.111	0.674	
Panel B. Predictive regression: Coefficient estimates on $\widehat{cay}^{top10}$				
	Fitted data: $\widehat{AS}, \widehat{YS}$		Raw data: AS, YS	
	WID	SZZ	WID	SZZ
NDS10	0.372* (1.723) [0.007]	0.369* (1.711) [0.007]	0.170 (1.419) [0.002]	0.221* (1.811) [0.007]
PCE10	0.629** (2.417) [0.020]	0.627** (2.403) [0.020]	0.165 (1.469) [0.003]	0.221* (1.915) [0.008]

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

7. Conclusion

In this paper we show that the ability of the consumption–wealth ratio, cay , to predict future market excess returns has dramatically declined over more than the last fifteen years in terms of in-sample, out-of-sample as well as economic performance. This decay of cay is most salient for the original construction of cay from NDS data, but also present when switching to PCE data, as suggested in Lettau and Ludvigson (2019). We investigate the reasons for the declining forecasting ability as well as potential remedies.

From an econometric perspective, we identify the source of the decay to be a structural shift in the cointegration relationship that governs the comovement between consumption and wealth. Economically speaking, asset wealth appears to be increasingly detached from its long-run relation with consumption (NDS as well as PCE) and labor income over at least the last two decades. This behavior implies that the cointegrating parameters of asset wealth and labor income are increasingly drifting apart. Among others, we show that this structural shift drives the poor OOS performance as well as the failure to generate abnormal returns over the past two decades. We do not take a stance on whether this behavior results from a long-lasting deviation from a previously reached equilibrium or convergence towards another equilibrium, which remains an open question for future research.

We offer a novel construction for cay , by taking up the argument put forward in, e.g., Lettau et al. (2019), that due to wealth inequalities, the representative agent may be better captured by the richest households. Therefore, we determine consumption and wealth data for the 10% of richest households. We find that the corresponding cay_t^{PCE10} is the dominant predictor and shows the weakest decline in forecasting power among a set of alternative approaches to improve cay . We therefore conclude that the most promising way forward to reconcile the poor empirical performance with the underlying theory is to identify the representative investor more carefully than by simply considering aggregate measures.

CRedit authorship contribution statement

Moritz Dauber: Writing – original draft, Data curation. **Jochen Lawrenz:** Writing – original draft, Supervision.

Appendix A. Construction of the top-10% consumption–wealth ratio

This section provides more information on the construction and the robustness of the top-10% consumption–wealth ratio cay^{top10} . The aggregate consumption, wealth and labor income series are adjusted for the top-10% richest households by multiplying them with the fitted values from the regressions described in Eqs. (7) and (8). First, we report coefficient estimates of these regressions in panel A of Table A.1. We provide results for two data-sources on wealth and income inequality: The data according to Saez and Zucman (2016), which is available through the World Inequality Data (WID), and the data according to Smith et al. (2023), which we label as SZZ. We report the temporal development of the initial raw data on wealth shares AS^{top10} , income share YS^{top10} , and capital share KS , as well as the fitted values $\widehat{AS}^{top10}$, $\widehat{YS}^{top10}$ within our sample period in Fig. A.1

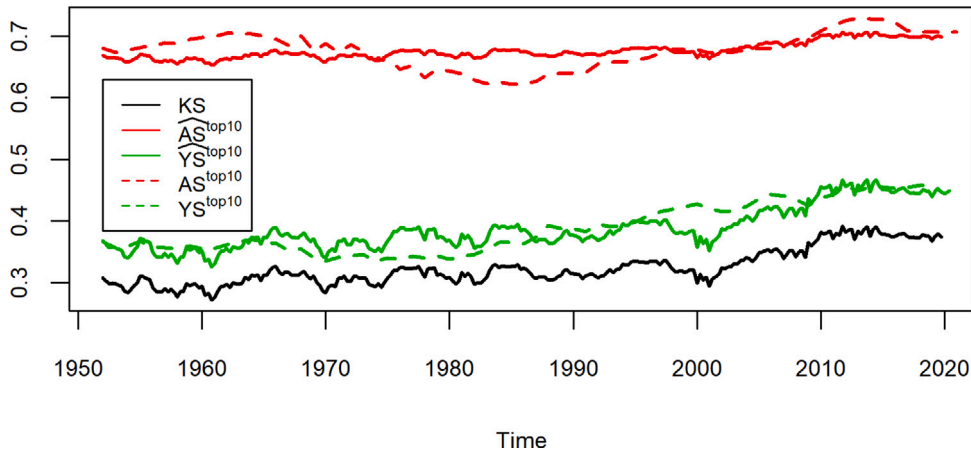


Fig. A.1. The plot shows the temporal development of the initial raw data on the wealth share, $\widehat{AS}^{\text{top10}}$, income share, $\widehat{YS}^{\text{top10}}$, and capital share, KS , as well as the fitted values $\widehat{AS}^{\text{top10}}$, $\widehat{YS}^{\text{top10}}$ within our sample period 1952:1–2019:4.

In panel B of [Table A.1](#), we report coefficient estimates on $\widehat{cay}^{\text{top10}}$ from the predictive regression according to Eq. (12). Rows 1 and 2 report results from using either NDS or PCE consumption adjusted for the top-10% share. In columns 1–2, we use fitted values for $\widehat{AS}$, $\widehat{YS}$ as in the main text. Columns 3–4 use the raw data AS , YS . Results with raw data are weaker overall, although pointing in the same direction, as results for SZZ and PCE10 are still significant and better than for the aggregate cay .

Using adjustment factors derived from fitted values of a regression on KS_t may raise the concern that this mechanically increases the stability of the cointegration relationship. The following formal derivation shows that this is not a necessary implication as the fitted values only act as scaling variables of the aggregate time-series. To see this, let $\tilde{A}_t$ and $\tilde{Y}_t$ be any two time series that can potentially be used to adjust C_t , A_t and Y_t for their top-10% share, such that $C_t^{\text{top10}} = C_t \cdot \tilde{Y}_t$, $Y_t^{\text{top10}} = Y_t \cdot \tilde{Y}_t$, and $A_t^{\text{top10}} = A_t \cdot \tilde{A}_t$. Then from the definition of cay , we have

$$cay_t^{\text{top10}} = c_t^{\text{top10}} - \alpha^{\text{top10}} a_t^{\text{top10}} - (1 - \alpha^{\text{top10}}) y_t^{\text{top10}},$$

where the superscript indicates the log consumption, wealth and income of the top 10% households. Plugging in and simplifying yields:

$$\begin{aligned} cay_t^{\text{top10}} &= \log(C_t \tilde{Y}_t) - \alpha^{\text{top10}} \log(A_t \tilde{A}_t) - (1 - \alpha^{\text{top10}}) \log(Y_t \tilde{Y}_t) \\ &= c_t + \log(\tilde{Y}_t) - \alpha^{\text{top10}} (a_t + \log(\tilde{A}_t)) - (1 - \alpha^{\text{top10}}) (y_t + \log(\tilde{Y}_t)) \\ &= c_t - \alpha^{\text{top10}} a_t - (1 - \alpha^{\text{top10}}) y_t + \alpha^{\text{top10}} \log\left(\frac{\tilde{Y}_t}{\tilde{A}_t}\right) \\ &= c_t - \alpha^{\text{top10}} (a_t - \psi_t) - (1 - \alpha^{\text{top10}}) y_t, \end{aligned} \quad (15)$$

where the last line uses the definition $\psi_t = \log\left(\frac{\tilde{Y}_t}{\tilde{A}_t}\right)$. This shows that ψ_t can be seen as an adjustment factor on aggregate wealth, which may or may not facilitate the initial cointegration relation.

We let ψ_t be the adjustment factor when using fitted values, i.e., by letting $\tilde{A}_t = \widehat{AS}_t^{\text{top10}} = \hat{\alpha}_A + \hat{\beta}_A KS_t$ and $\tilde{Y}_t = \widehat{YS}_t^{\text{top10}} = \hat{\alpha}_Y + \hat{\beta}_Y KS_t$. We further label ψ_t^{raw} as the corresponding adjustment from raw, unfitted data, i.e., $\psi_t^{\text{raw}} = \log\left(\frac{YS_t^{\text{top10}}}{AS_t^{\text{top10}}}\right)$. We provide evidence that the intermediate step of projecting on KS_t is able to improve the cointegration relation in an economically meaningful way as it captures relevant variation in shares of wealthy households. To this end, we repeat the analysis of Section 4.2 and report results on the cointegrating parameters as in [Fig. 7](#), but for the specification where the top 10%-adjustment is made from the raw time-series in [Fig. A.2](#). While [Fig. 7](#) in the main text shows that a stable cointegration relation is obtained by using fitted values, the usage of raw data (see [Fig. A.2](#)) is making the assumption of a stable cointegration worse than in the case of the aggregate cay .

Finally, we address the concern that the introduction of KS_t itself is driving the improved predictive power of cay . Therefore, we show results for predictive regressions from the aggregate cay to which we add KS_t or its first difference, ΔKS_t , as a second regressor. The results, reported in [Table A.2](#), show that neither KS_t nor ΔKS_t are significant while the coefficient estimates of cay_t remain qualitatively unaffected, thereby discarding the possible concern.

Appendix B. Data

Consumption. Consumption is defined in two different ways. One being nondurables plus services minus clothing and footwear (NDS). The other is total personal consumption expenditures (PCE). Data are quarterly, seasonally adjusted at an annual rate and

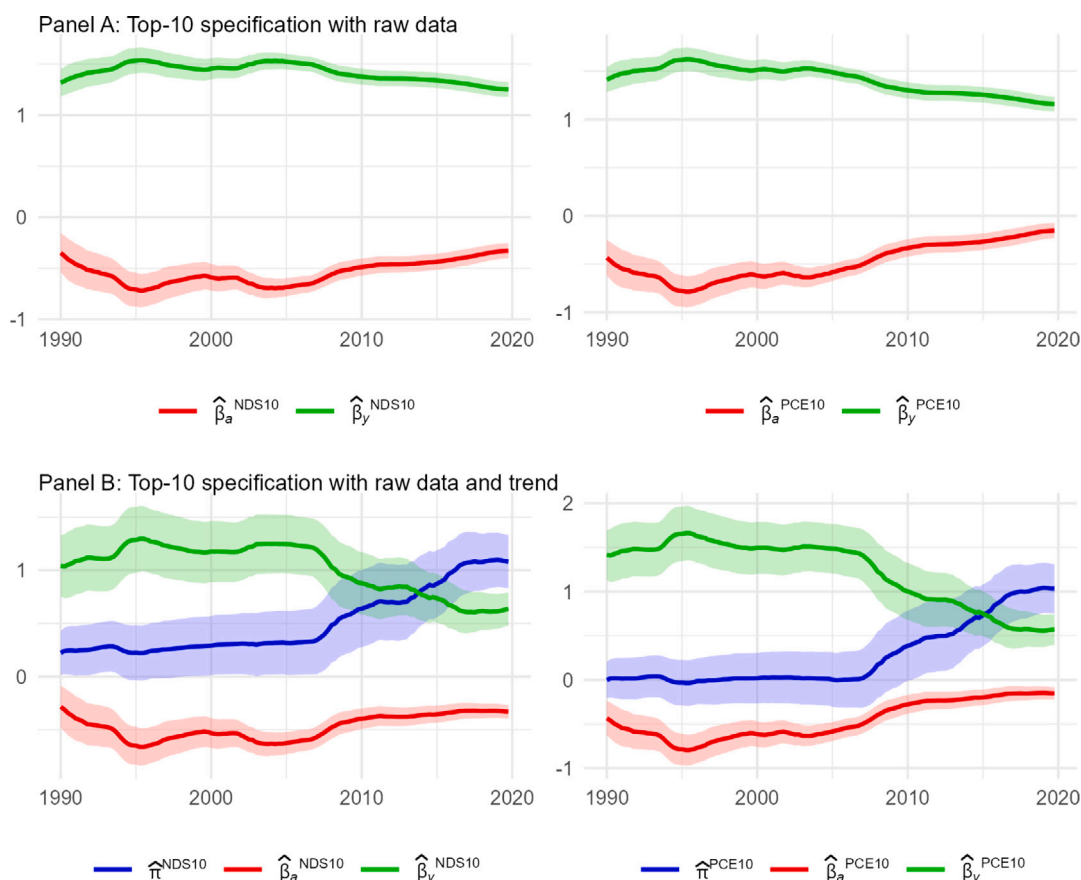


Fig. A.2. Cointegrating parameters for asset wealth (red), labor income (green) and a time trend (blue) quarterly re-estimated. The shaded areas represent the corresponding confidence bands of ± 2 standard errors. The sample period is 1952:1–2019:4.

Table A.2

Coefficient estimates of one-quarter ahead forecast regressions of market excess returns on $\widehat{cay}_t$ alone, and together with KS_t and ΔKS_t , respectively, for both NDS and PCE as the consumption measure. [Newey and West \(1987\)](#) corrected t -statistics appear in parentheses. The sample period is 1952:1–2019:4.

	NDS			PCE		
	(1)	(2)	(3)	(4)	(5)	(6)
$\widehat{cay}_t$	0.241 (1.118)	0.310 (1.379)	0.229 (1.062)	0.462* (1.698)	0.506* (1.823)	0.452* (1.697)
KS_t		0.232 (1.335)			0.215 (1.335)	
ΔKS_t			0.340 (1.417)			0.338 (1.432)
R^2	0.001	0.003	0.004	0.008	0.009	0.010

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

measured in billions of dollars. Series comprise the period 1947:1–2022:4. The source is the U.S. Bureau of Economic Analysis, NIPA Table 2.3.5, lines 1, 8, 10, 13.

Each series is separately deflated to 2017-dollars with its respective deflator, see below, before NDS is computed as the sum of nondurables and services minus clothing and footwear. Series are transformed in per capita terms and put in the logarithmic form.

Aggregate wealth. Aggregate wealth is defined as the net worth of households and nonprofit organizations. Data are quarterly, end of period, not seasonally adjusted. Series comprises the period 1951:4–2022:4. The source is the Board of Governors of the Federal Reserve System, Financial Accounts, Table B.101, line 40.

Financial wealth. Financial wealth is defined as financial assets (debt securities and loans, corporate equities, mutual fund shares, deposits, life insurance reserves, pension entitlements, miscellaneous assets, equity in noncorporate business, and grants and trade

Table C.1

Cointegrating parameters for standard OLS, Park's (1992) canonical cointegrating regression and Phillips and Ouliaris's (1990) fully modified estimation. Newey and West (1987)-corrected *t*-statistics appear in parentheses.

	NDS			PCE			NDS10			PCE10		
	OLS	CCR	FME	OLS	CCR	FME	OLS	CCR	FME	OLS	CCR	FME
$\hat{\beta}_a$	0.031 (0.957)	0.023 (0.704)	0.023 (0.708)	0.216*** (8.048)	0.215*** (8.202)	0.215*** (8.235)	0.069** (2.415)	0.067** (2.017)	0.067** (2.027)	0.219*** (9.281)	0.225*** (8.195)	0.225*** (8.221)
$\hat{\beta}_y$	0.922*** (25.972)	0.931*** (23.764)	0.931*** (23.760)	0.805*** (27.611)	0.805*** (26.050)	0.805*** (26.064)	0.893*** (32.907)	0.895*** (26.159)	0.895*** (26.148)	0.823*** (35.412)	0.817*** (28.708)	0.817*** (28.705)

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

receivables) minus financial liabilities (debt security and loans, trade payables, and deferred and unpaid life insurance premiums). Data are quarterly, end of period, not seasonally adjusted. Series comprise the period 1951:4–2022:4. The source is the Board of Governors of the Federal Reserve System, Financial Accounts, Table B.101, lines 9 and 30.

Housing wealth. Housing wealth is defined as the value of real estate held by households minus home mortgages. Data are quarterly, end of period, not seasonally adjusted. Series comprise the period 1951:4–2022:4. The source is the Board of Governors of the Federal Reserve System, Financial Accounts, Table B.101, lines 4 and 33.

Labor income. Labor income is defined as the sum of wages and salaries (line 3), personal current transfer receipts (line 16), and employer contributions for employee pension and insurance funds (line 7) minus employee contributions for government social insurance, and taxes. Employee contributions for government social insurance are defined as personal contributions for government social insurance (line 25) minus employer contributions for government social insurance (line 8). Taxes are defined as [(wages and salaries (line 3))/(wages and salaries (line 3) + proprietor's income with inventory valuation and capital consumption adjustments (line 9) + rental income of persons with capital consumption adjustment (line 12) + personal dividend income (line 15) + personal interest income (line 14))] * (personal current taxes (line 26)). Data are quarterly, seasonally adjusted at annual rates and measured in billions of dollars. Series comprise the period 1947:1–2022:4. The source is the U.S. Bureau of Economic Analysis, NIPA Table 2.1.

The aggregate wealth, financial wealth, housing wealth and labor income series are all deflated to 2017-dollars using the PCE chain-type price deflator before being transformed into per capita terms and put in the logarithmic form.

In the following, we will describe the population series as well as the price deflators used for that in more detail.

Population. Population is defined as the population series by the BEA, retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/B230RC0Q173SBEA>, April 5, 2023. Data are quarterly, not seasonally adjusted and comprise the period 1947:1–2022:4.

Price deflator. The chain-type price deflators for personal consumption expenditures, nondurables, services and clothing and footwear are quarterly, seasonally adjusted and in 2017-dollars. The source is the U.S. Bureau of Economic Analysis, NIPA Table 2.3.4, lines 1, 8, 10 and 13.

Inflation rate. The inflation rate is computed from the PCE chain-type price deflator. Series comprises the period 1947:2–2022:4.

Interest rate ("Risk-free" rate). The "risk-free" rate is defined as the 3-month U.S. Treasury bills real interest rate per quarter. Original data are monthly not seasonally adjusted nominal rates in percent per annum and are converted to a quarterly frequency by computing the simple arithmetic average of three consecutive months and applying the discount method. The real interest rates are computed as the difference between nominal interest rates and the inflation rate. The source of the 3-month U.S. Treasury bill secondary market rate is the H.15 publication of the Board of Governors of the Federal Reserve System and it comprises the period 1934:1–2022:4.

Asset returns. Asset returns are computed using the CRSP NYSE/NYSE MKT/NASDAQ/Arca Value-Weighted Market Index. It measures market performance assuming a full reinvestment of all distributions. The original series is daily. Quarterly returns are computed using the data from the last business day of each quarter. The series comprises the period 1926:1–2022:4.

Appendix C. Cointegration

C.1. Other cointegration estimates

Next to the DLS method used throughout the paper when estimating the cointegrating parameters of the long-run relationship between consumption, aggregate wealth and labor income, there are also other approaches of estimating said parameters. Table C.1 presents these for three other methods, namely OLS, Park's (1992) canonical cointegration regression (CCR) and Phillips and Ouliaris' (1990) fully modified estimator (FME).

We see that all methods produce similar cointegrating parameters (also to the DLS estimates from Table 1 and by that also almost identical forecasts.

Table C.2

The table reports test statistics from Engle and Granger (1987) cointegration tests of our four specifications over different sample periods.

Engle and Granger (1987) cointegration tests						
specification	90% CV	95%CV	1952:4-1998:3		1952:1-2019:4	
			Statistic	p-value	Statistic	p-value
NDS	-3.452	-3.743	-2.737	0.367	-1.556	0.884
PCE	-3.452	-3.743	-3.530	0.091	-2.054	0.579
NDS10	-3.452	-3.743	-2.753	0.359	-1.587	0.876
PCE10	-3.452	-3.743	-3.382	0.124	-2.659	0.404

Table C.3

The table reports Johansen (1988, 1991) test statistics for the aggregate NDS specification including one to four lags and assuming a linear trend in the underlying data but not in the cointegrating relationship itself. The sample period is 1952:1-2019:4.

L-Max		Trace		$H_0 = r$
Test Statistic	90% CV	Test Statistic	90% CV	r
Panel A: Lag in VAR Model equal to 1				
13.14	18.89	20.07	27.07	0
6.40	12.30	6.93	13.43	1
0.54	2.71	0.54	2.71	2
Panel B: Lag in VAR Model equal to 2				
13.01	18.89	20.33	27.07	0
6.91	12.30	7.32	13.43	1
0.41	2.71	0.41	2.71	2
Panel C: Lag in VAR Model equal to 3				
14.35	18.89	19.28	27.07	0
4.39	12.30	4.93	13.43	1
0.55	2.71	0.55	2.71	2
Panel D: Lag in VAR Model equal to 4				
20.69	18.89	25.12	27.07	0
4.18	12.30	4.43	13.43	1
0.26	2.71	0.26	2.71	2

C.2. Cointegration test results

In Section 4 we already reported results of Phillips and Ouliaris (1990) cointegration tests for two different sample periods. We augment these here with test results from Engle and Granger (1987) as well as Johansen (1988, 1991) tests in Tables C.2, C.3 and C.4. Note that the Johansen (1988, 1991) procedure is not a residual-based test of a single cointegration relation but rather also tests for the actual number of cointegration relationships. Since the underlying consumption, aggregate wealth and labor income series are all $I(1)$, the Johansen (1988, 1991) tests are valid. We allow for a linear trend in the data but no trend in the cointegration relationship that has an intercept.

Appendix D. Forecasting consumption growth

The theory centered around Eq. (2) also allows for the consumption-wealth ratio to predict future consumption growth rather than asset returns. This translates into the possibility of cay to predict Δc_{t+1} according to (5). To illustrate that this is still not the case, even though the predictability of asset returns has also vanished, we display the results from H -period ahead consumption growth forecasts on $\widehat{cay}_t$ for all four specifications in Table D.1. A notable exception is the case of cay^{PCE10} which displays forecastability with the economically intuitive negative sign, that is higher consumption relative to wealth today leads to lower future consumption growth. We discuss possible implications of this finding within Section 6 of the main text.

Appendix E. Strict out-of-sample predictability

See Fig. E.1.

Appendix F. Alternative specifications

In this appendix, we outline in detail suggestions to improve the forecasting ability of cay that we discuss in Section 5 of the paper. The proposed alterations include disaggregating wealth as in Sousa (2010), adding volatility to the predictive regression as in Guo (2006) and unfiltering consumption data as in Kroencke (2017).

Table C.4

The table reports Johansen (1988, 1991) test statistics for the top-10 PCE specification including one to four lags and assuming a linear trend in the underlying data but not in the cointegrating relationship itself. The sample period is 1952:1-2019:4.

L-Max		Trace		$H_0 = r$
Test Statistic	90% CV	Test Statistic	90% CV	r
Panel A: Lag in VAR Model equal to 1				
8.95	18.89	14.93	27.07	0
5.95	12.30	5.98	13.43	1
0.03	2.71	0.03	2.71	2
Panel B: Lag in VAR Model equal to 2				
9.77	18.89	16.83	27.07	0
7.03	12.30	7.06	13.43	1
0.03	2.71	0.03	2.71	2
Panel C: Lag in VAR Model equal to 3				
11.28	18.89	18.44	27.07	0
7.13	12.30	7.16	13.43	1
0.03	2.71	0.03	2.71	2
Panel D: Lag in VAR Model equal to 4				
11.99	18.89	18.64	27.07	0
6.64	12.30	6.65	13.43	1
0.01	2.71	0.01	2.71	2

Table D.1

The table reports coefficient estimates for all four specifications of $\widehat{cay}_t$ from forecasts of H -period ahead consumption growth. The forecasts span different horizons from one quarter up to five years. Newey and West (1987) corrected t -statistics appear in parenthesis and adjusted R^2 statistics in square brackets. The sample period is 1952:1-2019:4.

Dependent variable: $\Delta c_{t+1} + \dots + \Delta c_{t+H}$							
	$H = 1$	$H = 2$	$H = 4$	$H = 8$	$H = 12$	$H = 16$	$H = 20$
$\widehat{cay}_t^{\text{NDS}}$	0.005 (0.275) [−0.003]	0.012 (0.267) [−0.002]	0.027 (0.220) [−0.001]	0.040 (0.186) [−0.002]	0.062 (0.250) [−0.001]	0.096 (0.372) [0.000]	0.114 (0.386) [0.000]
$\widehat{cay}_t^{\text{PCE}}$	−0.032 (−1.054) [0.004]	−0.039 (−0.438) [0.001]	−0.068 (−0.312) [0.001]	−0.115 (−0.313) [0.001]	−0.175 (−0.466) [0.003]	−0.244 (−0.469) [0.006]	−0.369 (−0.724) [0.014]
$\widehat{cay}_t^{\text{NDS } 10}$	−0.011 (−0.247) [−0.004]	0.018 (0.224) [−0.003]	−0.021 (−0.091) [−0.004]	−0.0139 (−0.311) [0.000]	−0.204 (−0.486) [0.002]	−0.206 (−0.485) [0.002]	−0.222 (−0.472) [0.001]
$\widehat{cay}_t^{\text{PCE } 10}$	−0.139** (−2.137) [0.014]	−0.193 (−1.547) [0.014]	−0.411 (−1.212) [0.032]	−0.767 (−1.410) [0.072]	−1.238** (−2.231) [0.141]	−1.596*** (−3.882) [0.209]	−1.841*** (−4.634) [0.238]

Note: *p<0.1; **p<0.05; ***p<0.01.

F.1. Disaggregating wealth into financial and housing wealth

Sousa (2010) proposes to disaggregate wealth, meaning to split up (aggregate) asset wealth into (aggregate) financial wealth and (aggregate) housing wealth. In doing this, Sousa (2010) also questions the suitability of *aggregate* asset wealth as the optimal variable for the estimation of *cay* but contrary to our top-10 specification not from the standpoint of wealth inequalities but instead because he believes transitory movements in asset wealth are rather to be found in financial than in housing wealth. In a sample ending with the last quarter of 2008, he shows a stronger performance of his variable labeled *cday* compared to *cay*.

We have seen the decay of *cay* intensify after the global financial crisis in several figures by now, cf. Fig. 3, so it is a natural exercise to assess the development of *cday* over the course of our full sample. We again employ both NDS and PCE as consumption measures. The estimated consumption–wealth ratios are then given by

$$\widehat{cday}_t^{\text{NDS}} = c_t^{\text{NDS}} + 0.022 f_t - 0.094 u_t - 0.873 y_t, \quad (16)$$

(−1.214) (5.851) (29.621)

$$\widehat{cday}_t^{\text{PCE}} = c_t^{\text{PCE}} - 0.113 f_t - 0.116 u_t - 0.797 y_t, \quad (17)$$

(6.131) (8.185) (30.175)

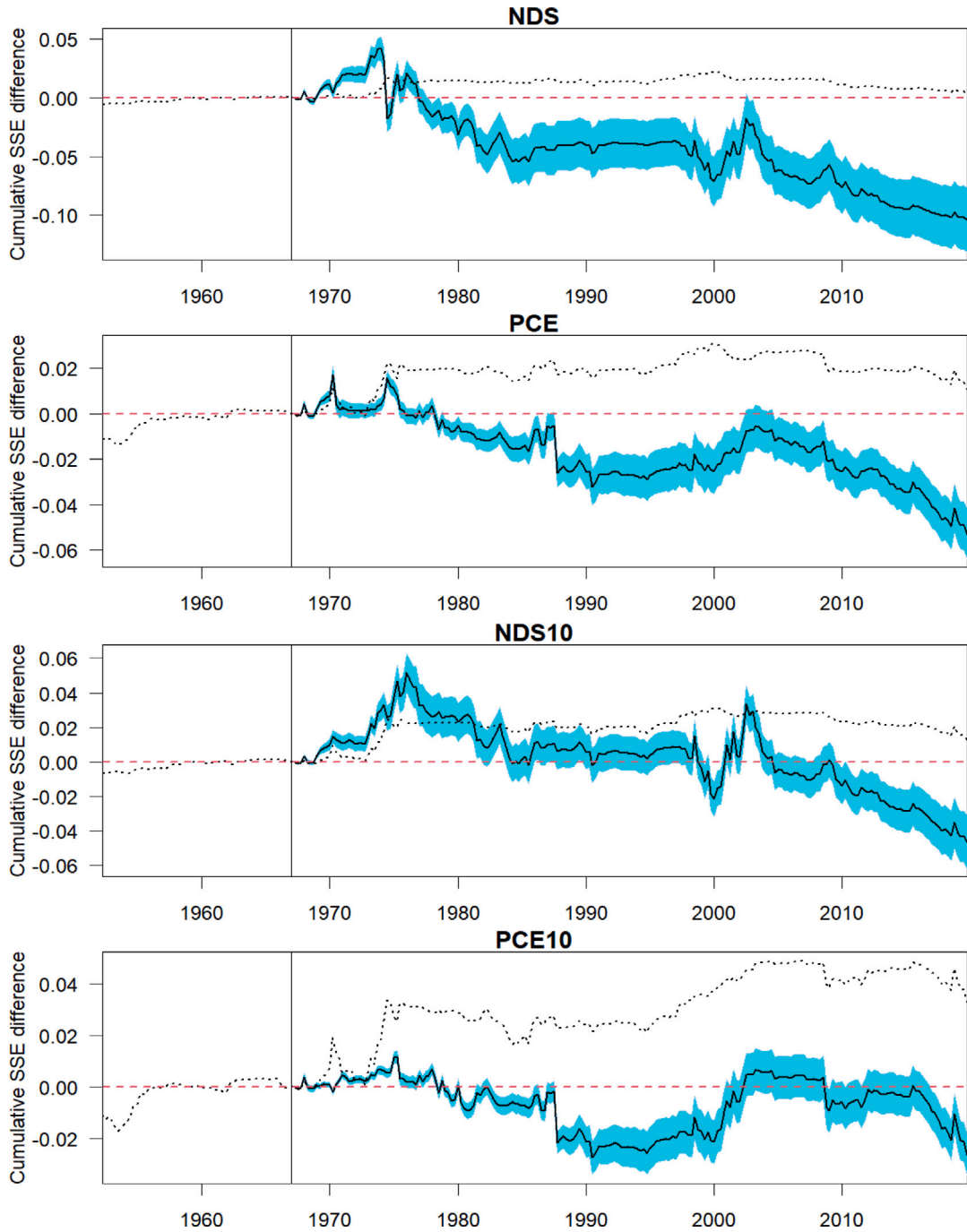


Fig. E.1. The plots show the OOS performance (solid) and the IS performance (dotted) of quarterly forecasts based on the respective *cay* in the spirit of [Welch and Goyal \(2008\)](#). Specifically, they show the cumulative SSE of a prevailing equity premium mean model minus the cumulative SSE of quarterly re-estimated *cay* forecasts for the OOS performance, and the difference between the cumulative SSE of the full sample equity premium mean and the cumulative SSE of forecasts based on *cay*, estimated over the full sample, for the IS performance. The blue 95%-confidence bands are computed using MSE-T critical values from [McCracken \(2007\)](#).

where f_t denotes financial wealth and u_t denotes housing wealth. The estimates from the forecasting regression

$$\tilde{r}_{t,H} = \alpha_{cday}^k + \beta_{cday}^k \widehat{cd ay}_t^k + \varepsilon_{t,H}^k, \quad k \in \{NDS, PCE\} \quad (18)$$

Table F.1

The table reports forecasting regression estimates for $\widehat{cday}_t^{NDS}$ and $\widehat{cday}_t^{PCE}$. Forecasts span different horizons from one quarter up to five years. Newey and West (1987) corrected t -statistics appear in parenthesis and adjusted R^2 statistics in square brackets. The sample period is 1952:1–2019:4.

Dependent variable: $\bar{r}_{t,H}$							
	$H = 1$	$H = 2$	$H = 4$	$H = 8$	$H = 12$	$H = 16$	$H = 20$
$\widehat{\beta}_{cday}^{NDS}$	0.285 (1.259) [0.001]	0.553 (1.153) [0.005]	0.981 (0.806) [0.010]	1.853 (0.726) [0.023]	2.402 (0.563) [0.030]	3.021 (0.598) [0.042]	2.925 (0.590) [0.030]
$\widehat{\beta}_{cday}^{PCE}$	0.497* (1.693) [0.009]	1.069* (1.806) [0.022]	2.373* (1.738) [0.058]	4.983** (2.272) [0.137]	6.553** (2.552) [0.174]	7.681*** (2.724) [0.201]	7.765** (2.401) [0.162]

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

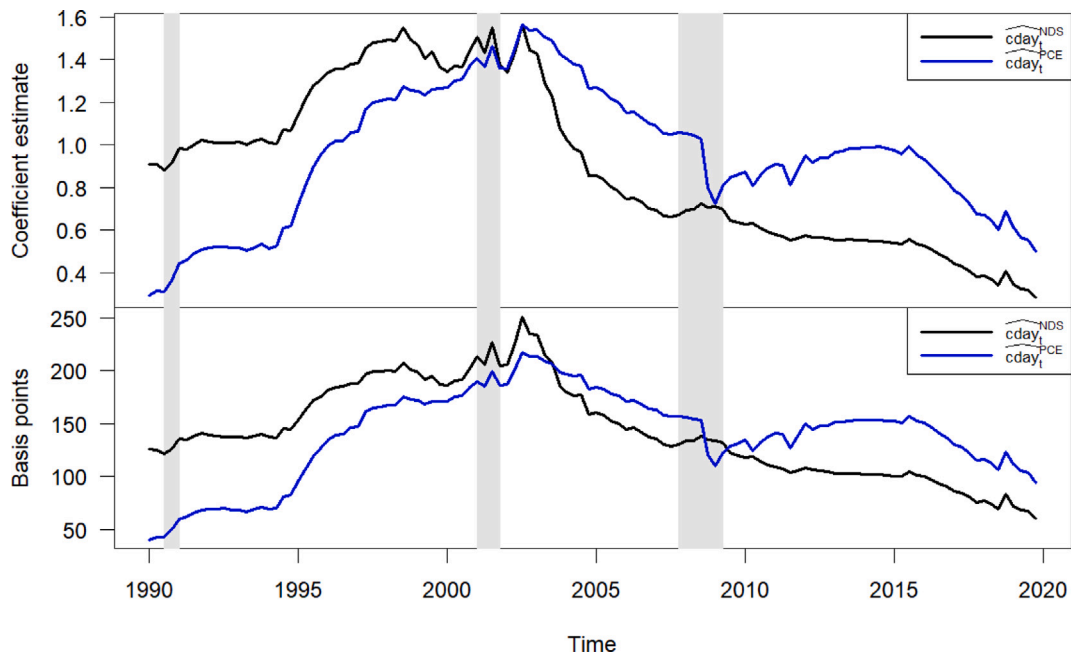


Fig. F.1. The plot in the upper panel shows the quarterly re-estimated coefficient estimate $\widehat{\beta}_{cday}$ of one-quarter ahead excess returns. The results for NDS as the consumption measure are shown in black, the ones for PCE in blue. The sample period is 1952:1–2019:4. The gray vertical bars mark NBER recessions. Likewise, the lower panel shows the increase of one-quarter ahead excess returns in basis points given a one standard deviation increase in the respective $\widehat{cday}_t$.

are shown in Table F.1. They are very close to their classical counterparts $\widehat{\beta}_{cay}^{NDS}$ and $\widehat{\beta}_{cay}^{PCE}$ from Table 3 and thereby, consistent with the findings of Sousa (2010), always a bit stronger. Conversely, this implies that $cday$ exhibits virtually the same decay as cay as can be seen in Fig. F.1.

F.2. Controlling for volatility in the forecasting regression

Guo (2006) argues that the forecasts of quarterly-ahead excess returns based solely on cay suffer from an omitted variables problem. He includes aggregate stock market volatility, σ_m^2 , into the forecast regression to get much stronger results for both the coefficient estimate of cay as well as the explained variation in terms of R^2 statistics. He mainly attributes this finding to the fact that while cay and σ_m^2 are negatively correlated to one another, they are both positively correlated with future stock returns. He furthermore shows that including the stochastically detrended risk-free rate, labeled $rrel$, into the forecast regressions yields significant estimates for all three variables and leads to an even higher R^2 statistic of 16.3% for a sample period of 1952:3–2002:4. These findings are much stronger when taking only the first half of the sample into account. Especially the finding that $rrel$ has some further explanatory power does not hold in the second half of the sample, whereas the omitted variable problem due to excluding σ_m^2 seems to still exist.

We address the omitted variables problem and follow Guo's (2006) approach by adding the variance of our market portfolio to the forecast regressions. We compute σ_m^2 by taking the sum of the squared daily log returns of the CRSP NYSE/NYSE

Table F.2

The table reports forecasting regression estimates for $\widehat{cay}_t^{\text{NDS}}$ and $\sigma_{m,t}^2$. Forecasts span different horizons from one quarter up to five years. Newey and West (1987) corrected t -statistics appear in parenthesis and adjusted R^2 statistics in square brackets. The sample period is 1952:1–2019:4.

Panel A: NDS consumption							
	Dependent variable: $\hat{r}_{i,H}$						
	$H = 1$	$H = 2$	$H = 4$	$H = 8$	$H = 12$	$H = 16$	$H = 20$
$\widehat{cay}_t^{\text{NDS}}$	0.240 (1.112)	0.455 (1.034)	0.798 (0.691)	1.510 (0.677)	2.125 (0.673)	2.932 (0.980)	3.428 (0.996)
$\sigma_{m,t}^2$	0.181 (0.219)	1.223 (1.447)	1.899 (1.531)	3.001 (1.546)	1.906 (0.854)	2.908 (1.374)	5.300** (2.540)
$\bar{R}^2$	−0.002	0.010	0.016	0.032	0.033	0.059	0.080
Panel B: PCE consumption							
	Dependent variable: $\hat{r}_{i,H}$						
	$H = 1$	$H = 2$	$H = 4$	$H = 8$	$H = 12$	$H = 16$	$H = 20$
$\widehat{cay}_t^{\text{PCE}}$	0.458* (1.732)	0.926* (1.767)	2.048 (1.432)	4.289* (1.877)	5.864** (2.228)	7.014*** (3.539)	7.530*** (3.112)
$\sigma_{m,t}^2$	0.100 (0.124)	1.056 (1.329)	1.528 (1.156)	2.206 (1.136)	0.902 (0.416)	1.728 (0.730)	4.045* (1.814)
$\bar{R}^2$	0.004	0.023	0.054	0.127	0.163	0.201	0.198

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

MKT/NASDAQ/Arca Value-Weighted Market Index for each quarter. The results for aggregate NDS and PCE are presented in Table F.2. We also included the relative bill rate, i.e., $rrel$, into the forecasts but did not find any significant results for $rrel$ itself, or any sizable improvement in R^2 statistics. Therefore, we decided to only incorporate σ_m^2 into our forecasts for the remainder of this section.¹⁹

In principal, the results including stock market volatility into the forecast regressions are just as bad as the ones for only using cay as a predictor variable as can be seen in Fig. F.2. This at first sight contrary finding to Guo (2006), highlights that the decay of cay is still present even when adding stock market volatility to our forecast regressions.

F.3. Unfiltering consumption data

Another attempt to improve the measurement of the consumption–wealth ratio is inspired by the work of Kroencke (2017). Picking up an issue raised by Savov (2011), namely that the published consumption series from the BEA are subsequently smoothed (or filtered) and thus may lose important information from an asset pricing perspective, Kroencke (2017) develops a model to undo the filtering procedure applied by the BEA.

Instead of simply using the typical approach of introducing new preferences to solve the standard equity premium puzzle of Mehra and Prescott (1985), Savov (2011) considers an alternative measure of consumption than the typically used nondurables and services (NDS) from the NIPA tables by the BEA. His measure, municipal solid waste (MSW), or simply garbage, is two and a half times more volatile and more highly correlated with stock returns. He then goes on to conclude that it is in fact the only measure able to resolve the joint equity premium-risk free rate puzzle of Weil (1989).

Arguing in the same direction, Kroencke (2017) develops a model of the filtering process by the BEA in order to undo the filtering. This process aims at getting back the original, more volatile consumption series. In his baseline method he uses a simple GARCH-process to model consumption volatility. The respective specifications of cay resulting from using unfiltered consumption are denoted as cay^{unfil} .

Appendix G. Multivariate predictive regressions

Confronted with the criticism that the construction of cay as a cointegrating residual involves the alleged ‘look-ahead’ bias, Lettau and Ludvigson (2005a) argue that one can also avoid this two-step approach and directly estimate a multivariate regression of excess returns on consumption, aggregate wealth and labor income. This procedure is valid under the assumption that the series

¹⁹ Different from Guo (2006), we did not adjust downward observations with an unusually high amplitude of realized stock market variance. The main reason for that is, next to the somewhat arbitrary decision which observations should be adjusted downward, that our findings are simply not sensitive to downward adjustment. By that we mean that we indeed do get statistically stronger results when adjusting downward, but our main interest lies in the decay of cay which is given in almost the same way, independent of downward adjustment. We therefore spare ourselves the decision which observations should be adjusted downward.

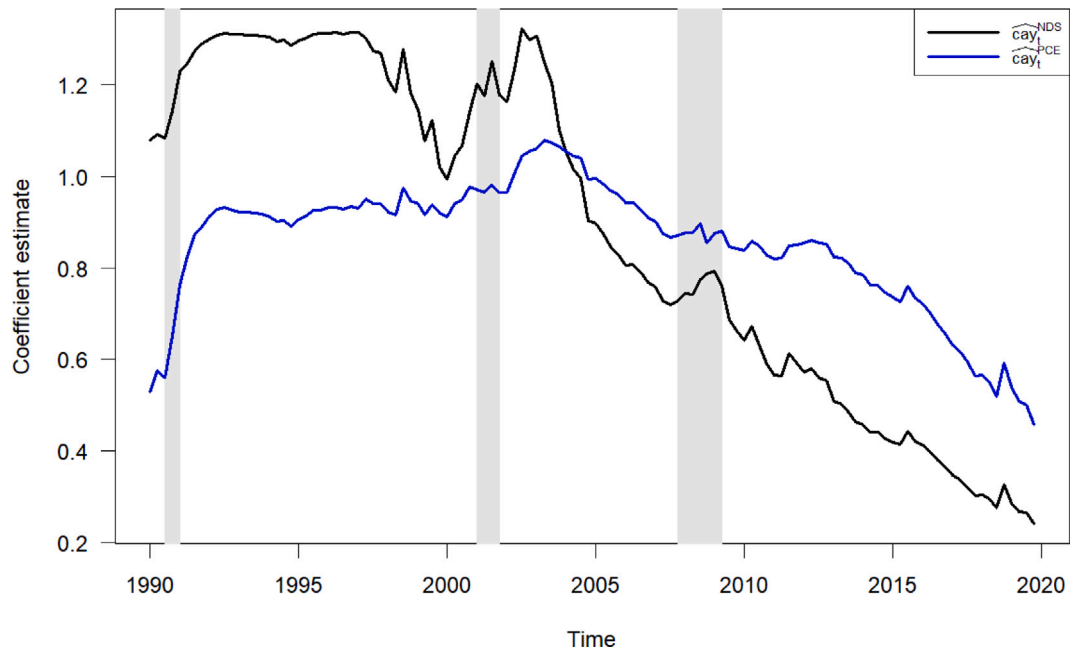


Fig. F.2. The plot shows the coefficient estimates of $\widehat{cay}_t$ for NDS (black) and PCE (blue) from quarterly forecast regressions of excess returns that also include the realized stock market variance $\sigma_{m,t}^2$. The sample period is 1952:1–2019:4.

Table G.1

The table reports regression estimates for c_t^{NDS} , a_t and y_t and adjusted R^2 statistics from forecasts of H -period ahead excess returns spanning different horizons from one quarter up to five years. Newey and West (1987) corrected t -statistics appear in parenthesis. The sample period is 1952:1–2019:4.

Dependent variable: $\tilde{r}_{t,H}$							
	$H = 1$	$H = 2$	$H = 4$	$H = 8$	$H = 12$	$H = 16$	$H = 20$
c_t^{NDS}	0.240 (1.104)	0.467 (1.061)	0.850 (0.728)	1.696 (0.769)	2.438 (0.722)	3.410 (0.797)	4.182 (1.057)
a_t	−0.014 (−0.306)	−0.051 (−0.569)	−0.156 (−0.565)	−0.313 (−0.470)	−0.418 (−0.487)	−0.529 (−0.418)	−0.712 (−0.872)
y_t	−0.217 (−0.963)	−0.397 (−0.904)	−0.654 (−0.557)	−1.312 (−0.621)	−1.917 (−0.626)	−2.730 (−0.812)	−3.266 (−0.966)
$\bar{R}^2$	−0.006	−0.001	0.008	0.029	0.044	0.068	0.078

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

are cointegrated and ensures that there are no future observations in use. Taking an agnostic stance on the underlying assumption of cointegration, we investigate whether the decay of cay , i.e., the disappearance of the forecasting ability of cay can also be found in the direct multivariate predictive regression. In line with Lettau and Ludvigson (2005a), we estimate

$$\tilde{r}_{t,H} = \alpha + \beta_c c_t + \beta_a a_t + \beta_y y_t + \varepsilon_{t,H} \quad (19)$$

for all four of our specifications. The results for horizons up to five years and all four specifications can be found in Tables G.1–G.4.

Clearly, there are no significant results for any of the three regressors over any horizon using aggregate NDS. Moreover, the coefficient estimates for consumption in the first row are very close to the ones for $\widehat{cay}_t$ from Table 3, especially for the shorter horizons. This is exactly what we would asymptotically expect in large samples. Taken together, the results in Tables G.1–G.4 document that from most recent data, future returns cannot be predicted over any horizon from the direct multivariate regression, except again for the top-10 specification using PCE.

Appendix H. The covid-19 pandemic episode

We deliberately ended our sample in 2019:4 and excluded observations that were impacted by the Covid-19 pandemic, hitting the U.S. in the first quarter of 2020. In this appendix, we illustrate the impact of the pandemic-affected observations on cay .

Table G.2

The table reports regression estimates for c_t^{PCE} , a_t and y_t and adjusted R^2 statistics from forecasts of H -period ahead excess returns spanning different horizons from one quarter up to five years. Newey and West (1987) corrected t -statistics appear in parenthesis. The sample period is 1952:1–2019:4.

	Dependent variable: $\tilde{r}_{t,H}$						
	$H = 1$	$H = 2$	$H = 4$	$H = 8$	$H = 12$	$H = 16$	$H = 20$
c_t^{PCE}	0.460 (1.649)	0.964* (1.893)	2.107 (1.452)	4.471* (1.786)	6.145 (1.645)	7.534** (2.142)	8.487*** (3.327)
a_t	-0.107* (-1.709)	-0.248* (-1.758)	-0.601* (-1.762)	-1.288 (-1.215)	-1.787 (-1.026)	-2.204 (-1.040)	-2.601** (-2.397)
y_t	-0.365 (-1.470)	-0.740* (-1.731)	-1.549 (-1.178)	-3.286 (-1.569)	-4.500* (-1.737)	-5.500*** (-2.744)	-6.060*** (-3.201)
$\bar{R}^2$	0.001	0.013	0.049	0.131	0.180	0.221	0.213

Note: *p<0.1; **p<0.05; ***p<0.01.

Table G.3

The table reports regression estimates for $c_t^{\text{NDS } 10}$, $a_t^{\text{top } 10}$ and $y_t^{\text{top } 10}$ and adjusted R^2 statistics from forecasts of H -period ahead excess returns spanning different horizons from one quarter up to five years. Newey and West (1987) corrected t -statistics appear in parenthesis. The sample period is 1952:1–2019:4.

	Dependent variable: $\tilde{r}_{t,H}$						
	$H = 1$	$H = 2$	$H = 4$	$H = 8$	$H = 12$	$H = 16$	$H = 20$
$c_t^{\text{NDS } 10}$	0.358 (1.565)	0.694 (1.475)	1.343 (1.105)	2.728 (1.271)	3.745 (1.318)	4.991 (1.640)	5.710** (2.060)
$a_t^{\text{top } 10}$	-0.081 (-1.501)	-0.171 (-1.362)	-0.392 (-1.287)	-0.785 (-0.948)	-0.984 (-1.196)	-1.188 (-1.294)	-1.298** (-2.004)
$y_t^{\text{top } 10}$	-0.264 (-1.198)	-0.499 (-1.163)	-0.906 (-0.798)	-1.865 (-0.972)	-2.659 (-1.041)	-3.665 (-1.403)	-4.254* (-1.681)
$\bar{R}^2$	0.002	0.013	0.042	0.101	0.124	0.165	0.156

Note: *p<0.1; **p<0.05; ***p<0.01.

Table G.4

The table reports regression estimates for $c_t^{\text{PCE } 10}$, $a_t^{\text{top } 10}$ and $y_t^{\text{top } 10}$ and adjusted R^2 statistics from forecasts of H -period ahead excess returns spanning different horizons from one quarter up to five years. Newey and West (1987) corrected t -statistics appear in parenthesis. The sample period is 1952:1–2019:4.

	Dependent variable: $\tilde{r}_{t,H}$						
	$H = 1$	$H = 2$	$H = 4$	$H = 8$	$H = 12$	$H = 16$	$H = 20$
$c_t^{\text{PCE } 10}$	0.611** (2.140)	1.215** (2.438)	2.579* (1.785)	5.331** (2.242)	6.920*** (2.643)	8.160*** (4.918)	8.404*** (4.058)
$a_t^{\text{top } 10}$	-0.191** (-2.538)	-0.391** (-2.274)	-0.876** (-2.413)	-1.797* (-1.745)	-2.288** (-2.144)	-2.661*** (-3.246)	-2.745*** (-3.515)
$y_t^{\text{top } 10}$	-0.447* (-1.790)	-0.878** (-2.163)	-1.822 (-1.374)	-3.792* (-1.886)	-4.976** (-2.288)	-5.902*** (-4.274)	-6.075*** (-3.542)
$\bar{R}^2$	0.013	0.035	0.097	0.233	0.284	0.328	0.271

Note: *p<0.1; **p<0.05; ***p<0.01.

Recall once more that a low cay is assumed to predict low future returns, since the representative agent reduces consumption in anticipation of lower growth rates for wealth. It is therefore instructive to first look at the development of the consumption and income time series during the Covid-19 episode, whose quarterly growth rates are shown in the upper two panels of Fig. H.1. It is more than obvious that the pandemic-affected quarters display extraordinary behavior. In 2020:1 and 2020:2, consumption growth dropped at an unprecedented magnitude while income spiked upwards in 2020:2 and 2021:1. A decline in consumption (i.e., the numerator) together with an increase in income (i.e., the denominator) propelled the consumption–wealth ratio eight standard deviations below its historic mean (see third panel in Fig. H.1). Within the interpretation of the theory, it would be a signal that the representative agent has massively reduced consumption because of anticipated extraordinarily low future returns. While it is true that the pandemic induced a negative shock on growth expectations, a closer inspection of the episode clearly shows that the

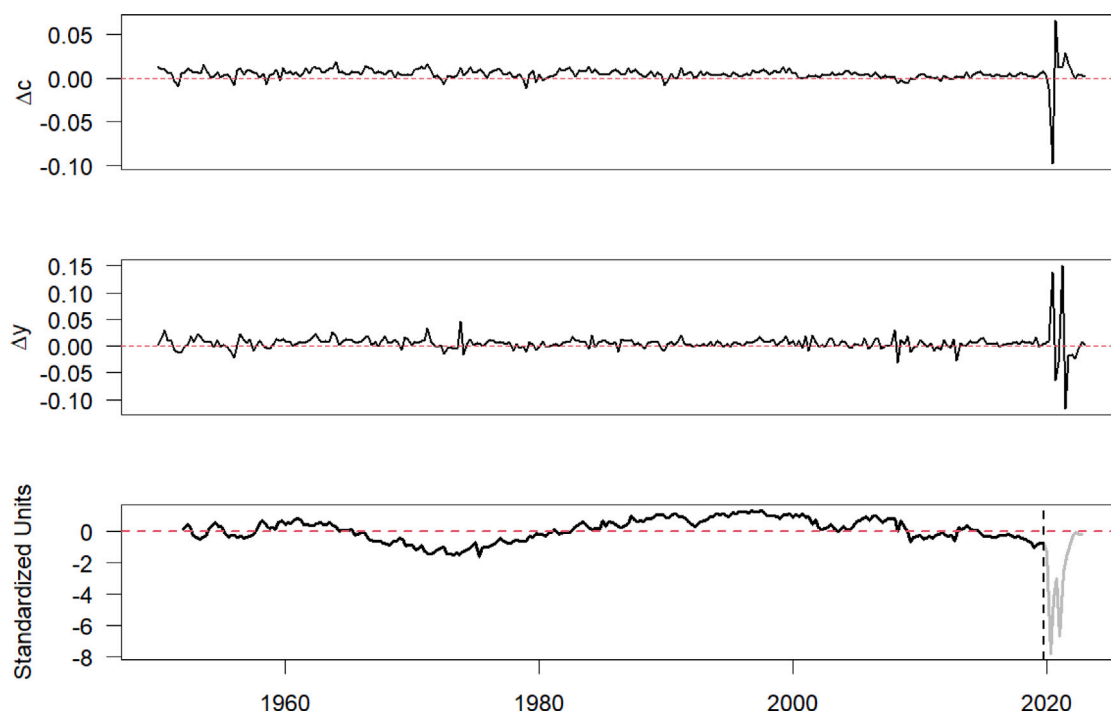


Fig. H.1. The two plots in the top show the (log) growth rates of consumption and labor income, respectively. The bottom plot shows $\widehat{cay}_t$ in standardized units. The sample period is 1952:1–2022:4.

Table H.1

The table reports forecasting regression estimates for $\widehat{cay}_t^{\text{NDS}}$ and $\widehat{cay}_t^{\text{PCE}}$. Forecasts span different horizons from one quarter up to five years. Newey and West (1987) corrected t -statistics appear in parentheses and adjusted R^2 statistics in square brackets. The sample period is 1952:1–2022:4.

	Dependent variable: $\bar{r}_{t,H}$						
	$H = 1$	$H = 2$	$H = 4$	$H = 8$	$H = 12$	$H = 16$	$H = 20$
$\widehat{\beta}_{cay}^{\text{NDS}}$	−0.063 (−0.377) [−0.003]	−0.101 (−0.288) [−0.003]	−0.049 (−0.062) [−0.003]	0.394 (0.317) [−0.000]	1.155 (0.337) [0.009]	1.864 (0.581) [0.024]	2.199 (0.644) [0.027]
$\widehat{\beta}_{cay}^{\text{PCE}}$	−0.012 (−0.053) [−0.004]	0.020 (0.047) [−0.004]	0.282 (0.293) [−0.001]	1.352 (1.090) [0.024]	3.738 (1.190) [0.082]	4.817* (1.698) [0.114]	5.203* (1.766) [0.105]

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

decline in consumption is primarily due to government-imposed shutdowns, while the huge spike in income can be explained by the substantial transfer payments that were made in the context of various rescue packages. Therefore, although the sign would be correct, it is hard to argue that during the Covid-19 episode cay is able to forecast future returns on the basis of its theoretical rationale. Consumption declined rather due to the inability to spend than due to anticipated low growth rates.

In sum, observations that are affected by the Covid-19 pandemic are subject to circumstances (economy shutdown, transfer payments) that are not captured by the theoretical model assumptions. Furthermore, by their extraordinary order of magnitude, these observations massively affect the proper estimation of $\widehat{cay}_t$. Unsurprisingly, this leads to a complete disappearance of virtually any stock return predictability as can be seen in Table H.1. Therefore, we limit ourselves to the sample ending in the fourth quarter of 2019.

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